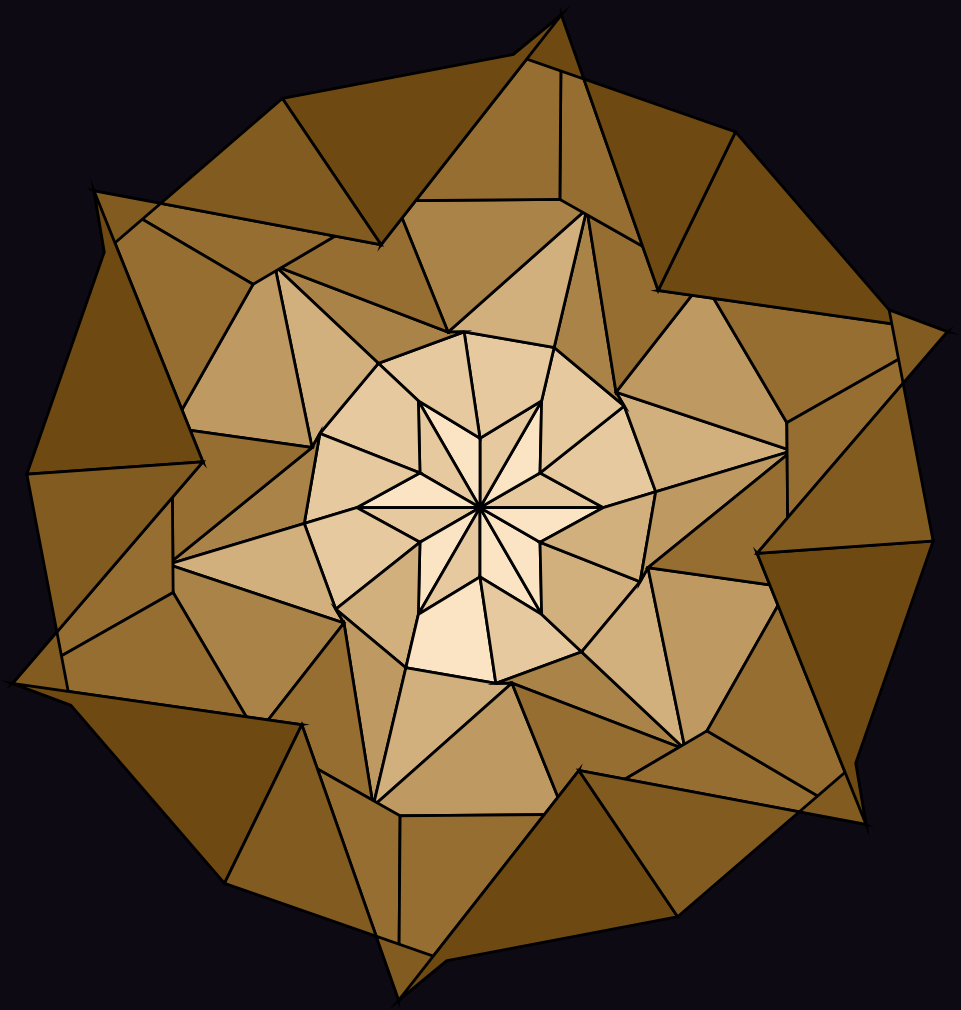


ISLAMIC STAR GEOMETRY

A Color-by-Number Journey

50 Geometric Star Rosette Designs

MathRelaxing

THE MATHEMATICS

Every rosette in this book is built the same way real Islamic geometric artisans laid out star patterns — with a compass and a grid of concentric circles, formalized here as an algorithm.

Points are placed on concentric rings around a center, evenly spaced by rotational symmetry of order n (8, 9, 10, 12 or 16-fold). Every other ring receives an alternating outward/inward radius shift, which turns a plain ring of points into a sharp n-pointed star. Consecutive rings are rotated by a small twist angle, so the bands between them read as an interlaced weave rather than a flat grid.

The resulting mesh of triangles and quadrilaterals is numbered by distance from the center — an outward gradient across 8 tones of a single hue, the same numbering convention used across every MathRelaxing color-by-number book.

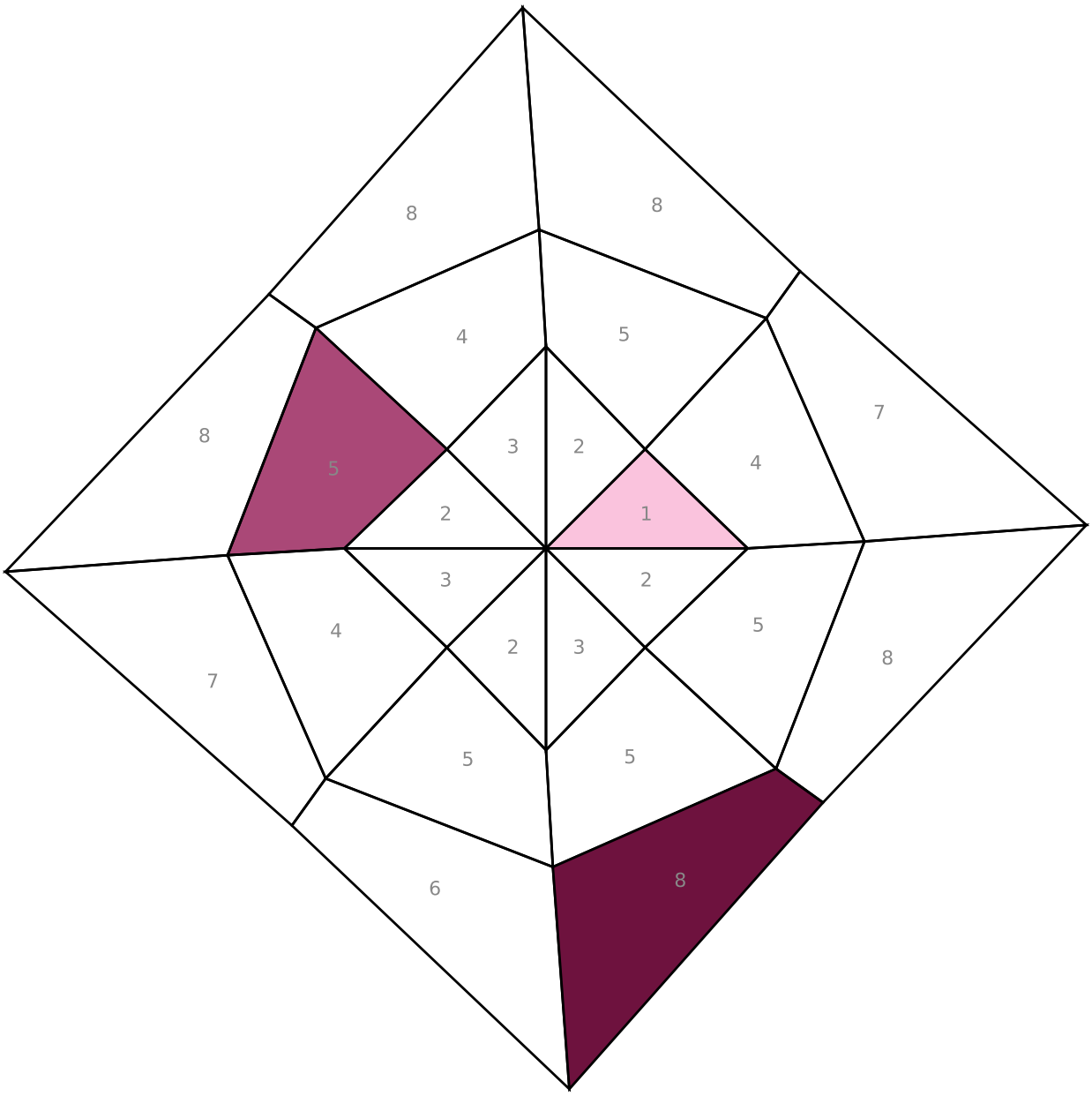
Ten star families appear — Octagram, Decagram, Dodecagram, Hexagram, Enneagram and more — each explored across five variations in ring count, twist and star sharpness.

© MathRelaxing. For personal use only. Not for redistribution or resale.

How to colour

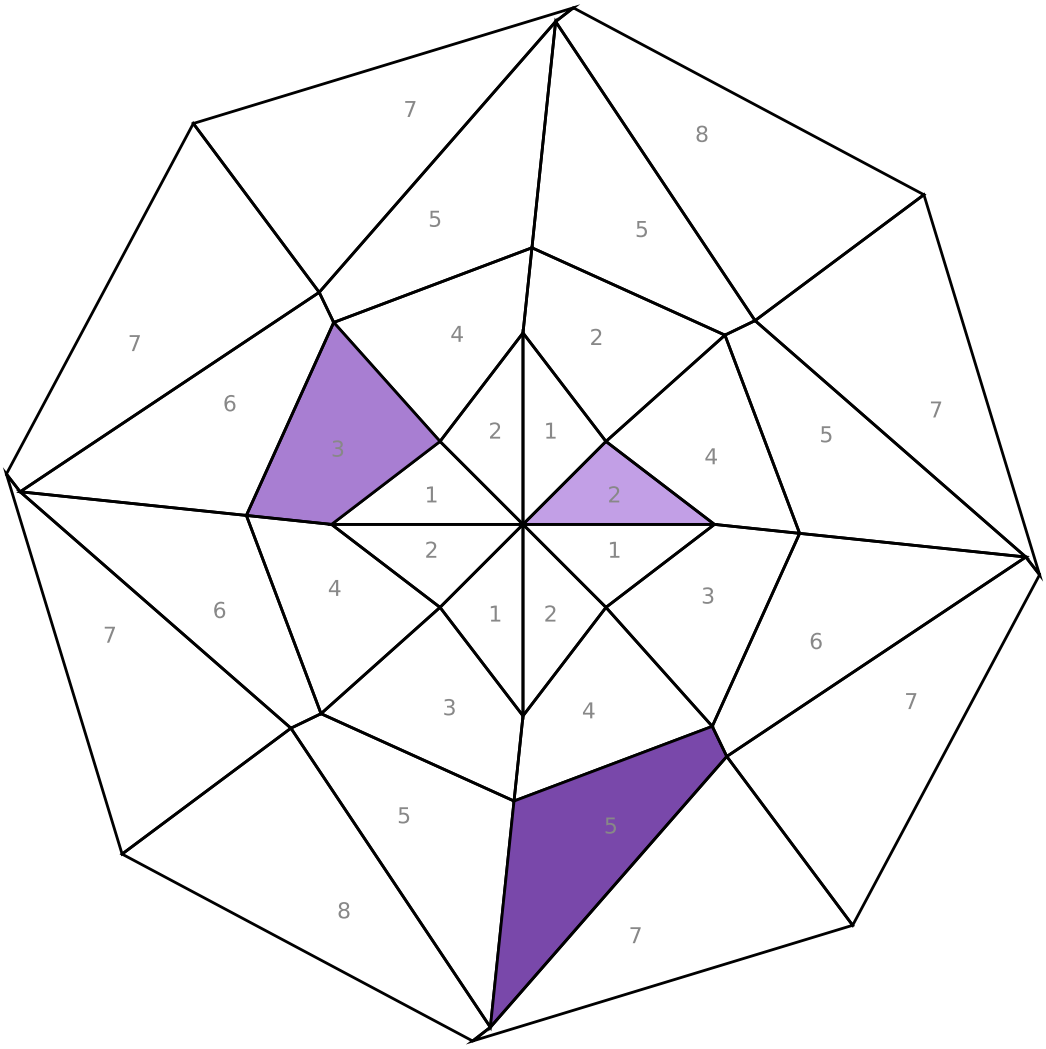
1. Match the numbers in the cells to the colour legend at the bottom of the page.
2. Relax and enjoy the star pattern emerge from flat facets!
3. A few accent cells have been pre-coloured for your convenience.

Use any pens, pencils, or markers. Slip a spare sheet behind the page if using wet markers.



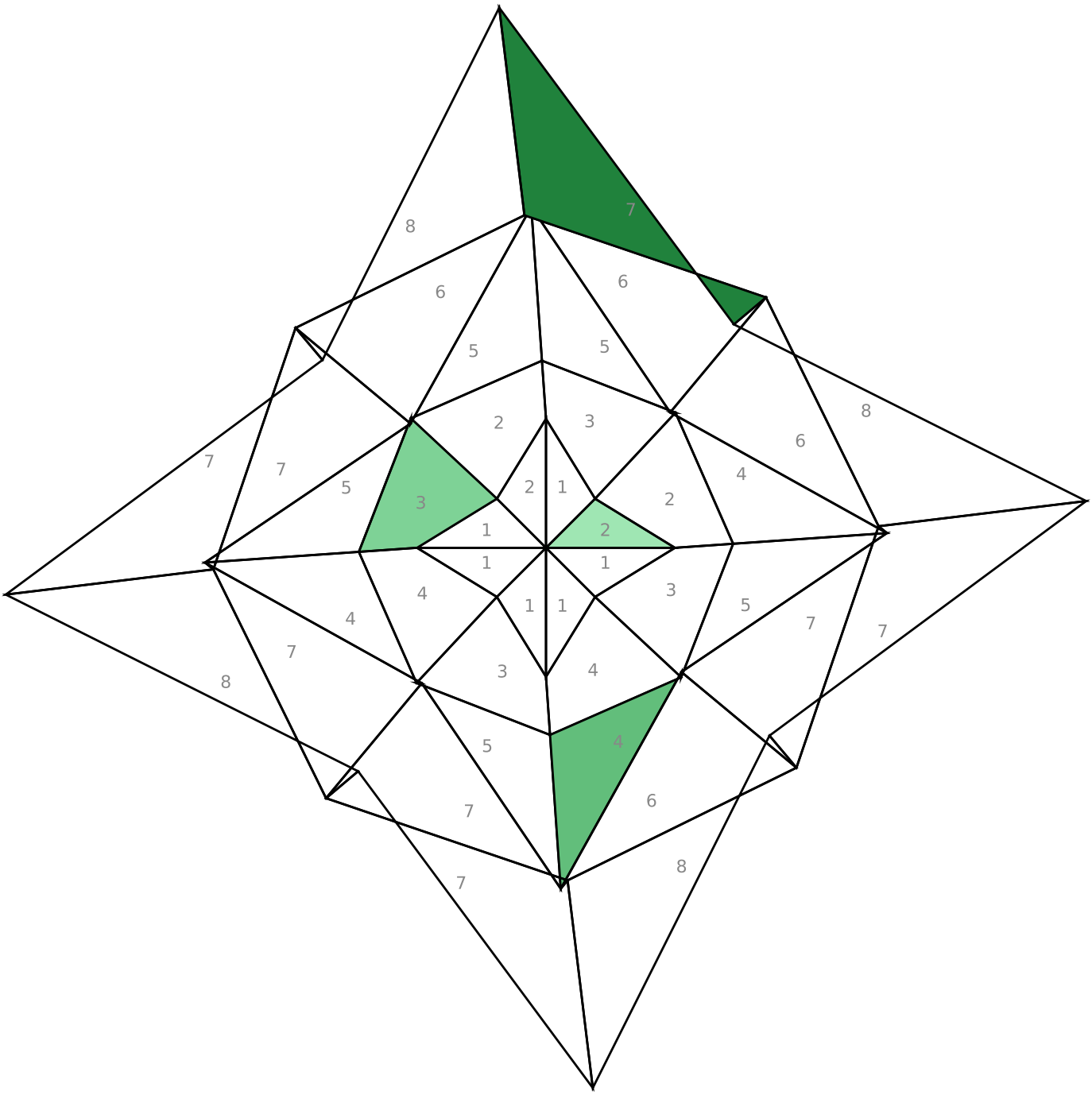
1. Octagram Rosette 1

1	2	3	4	5	6	7	8



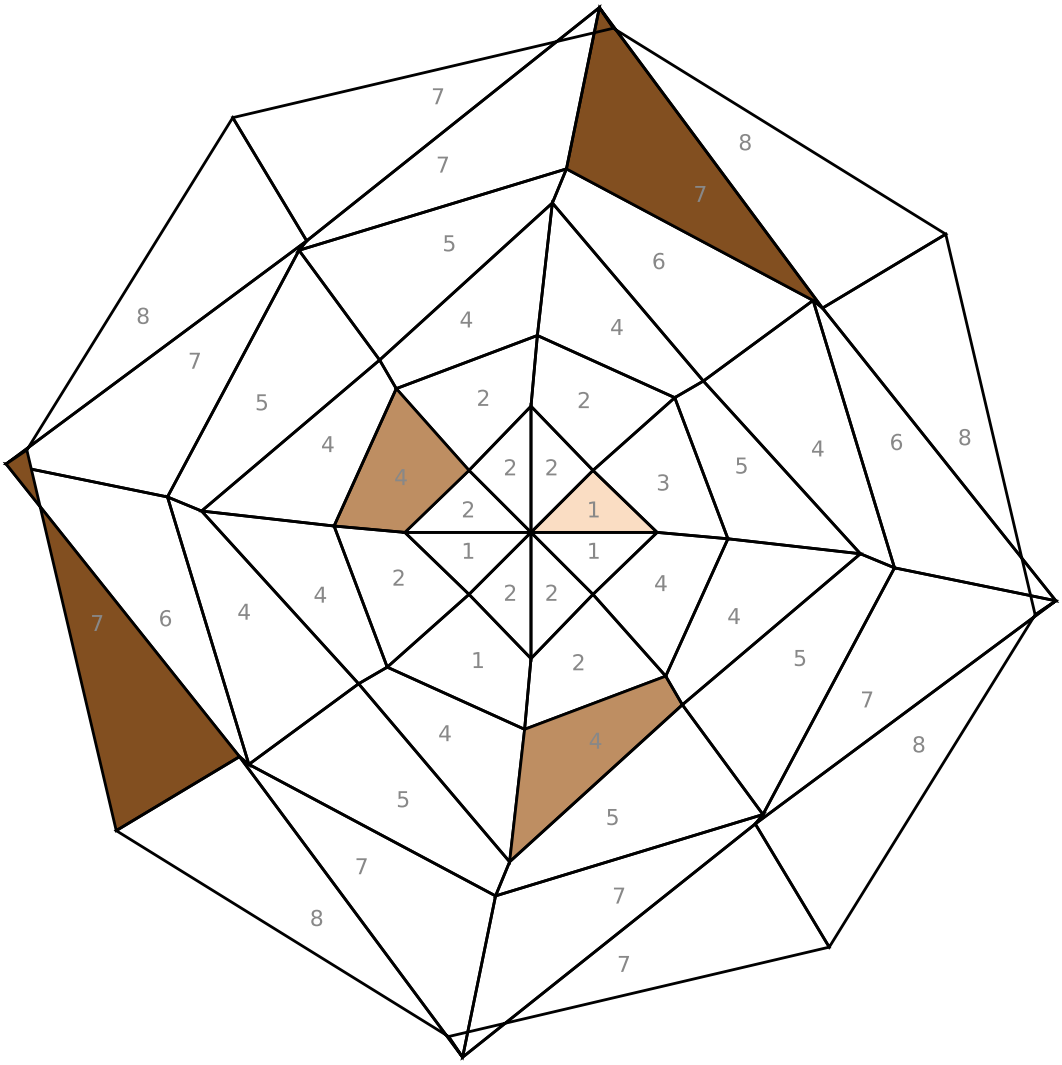
2. Octagram Rosette 2

1	2	3	4	5	6	7	8



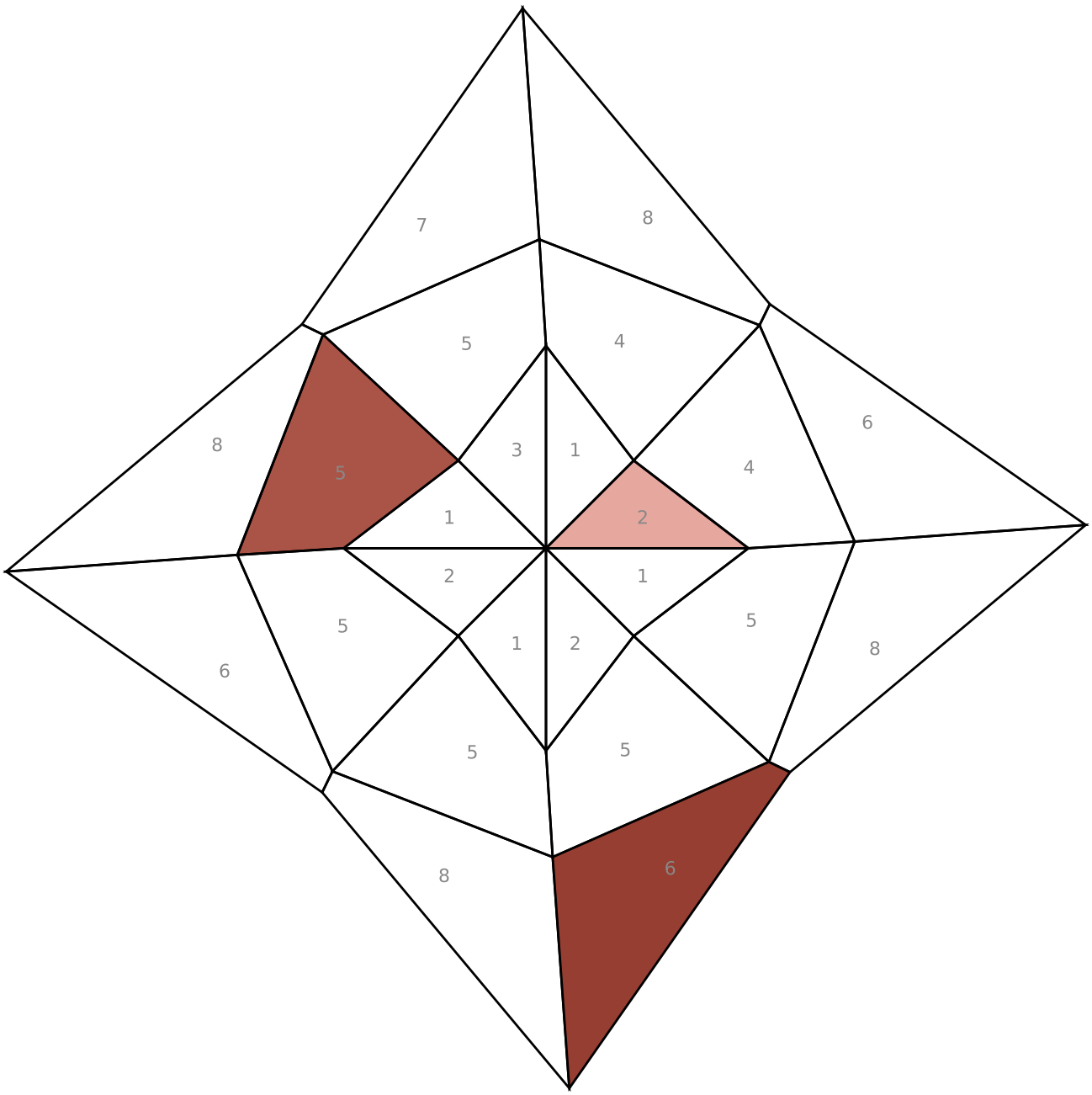
3. Octagram Rosette 3

1	2	3	4	5	6	7	8



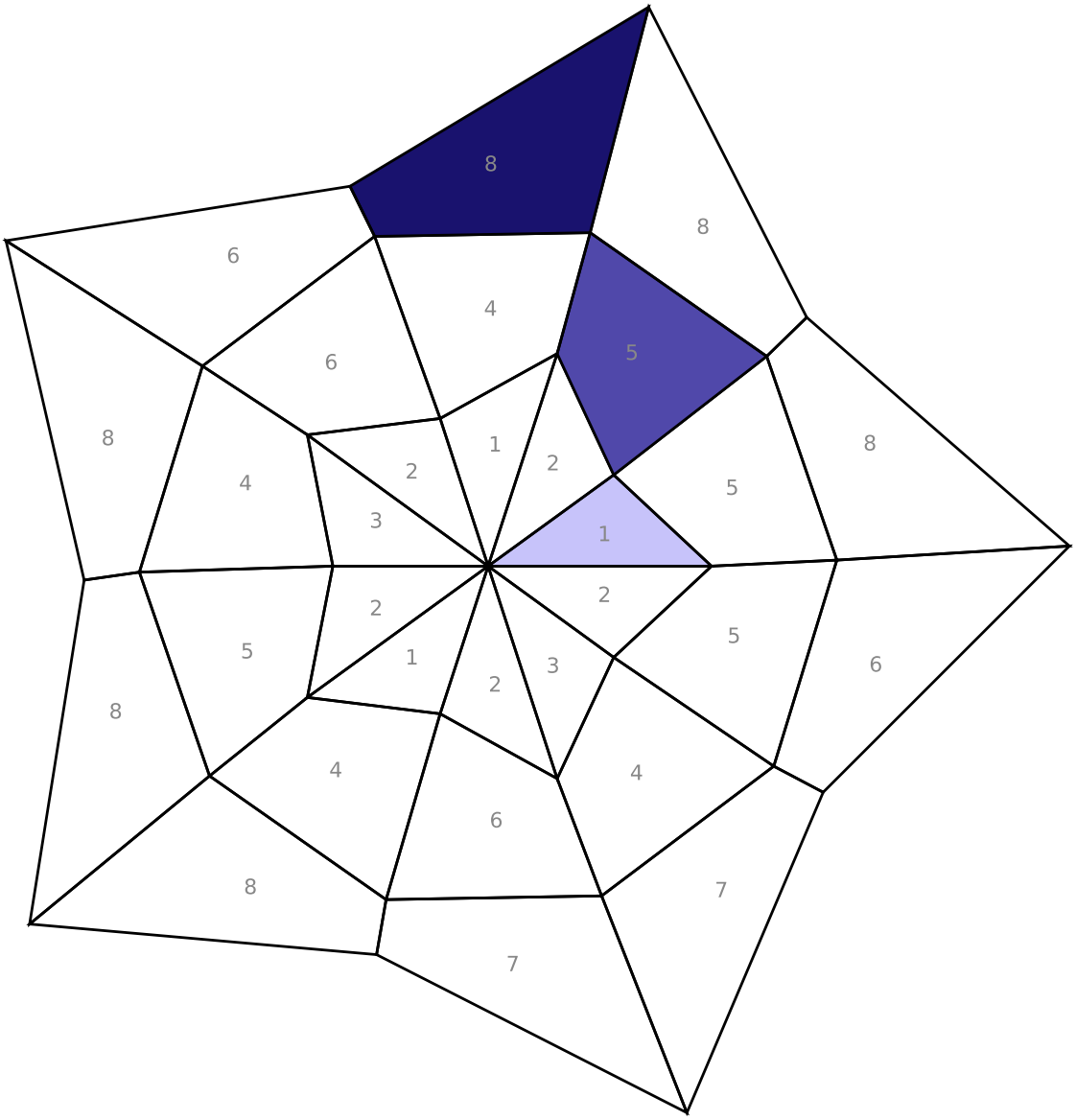
4. Octagram Rosette 4

1	2	3	4	5	6	7	8



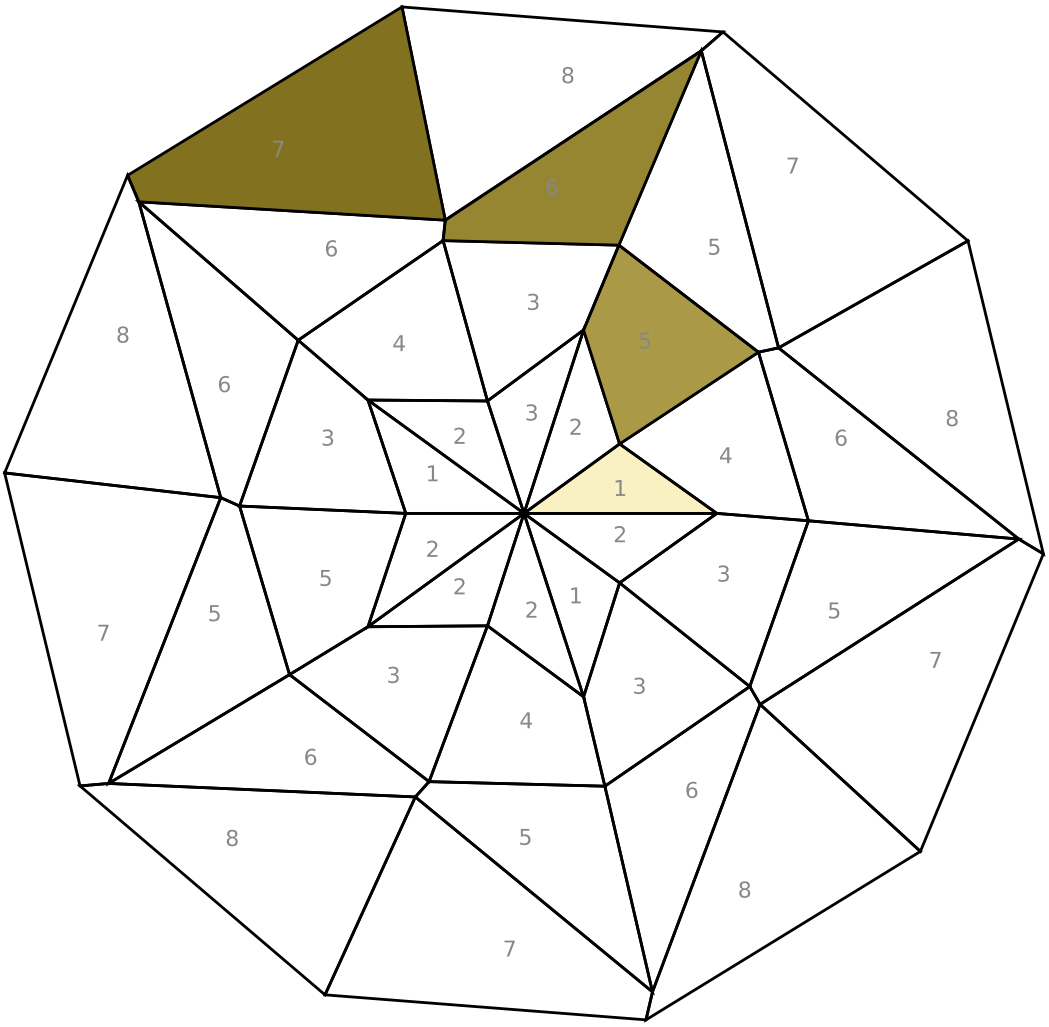
5. Octagram Rosette 5

1	2	3	4	5	6	7	8



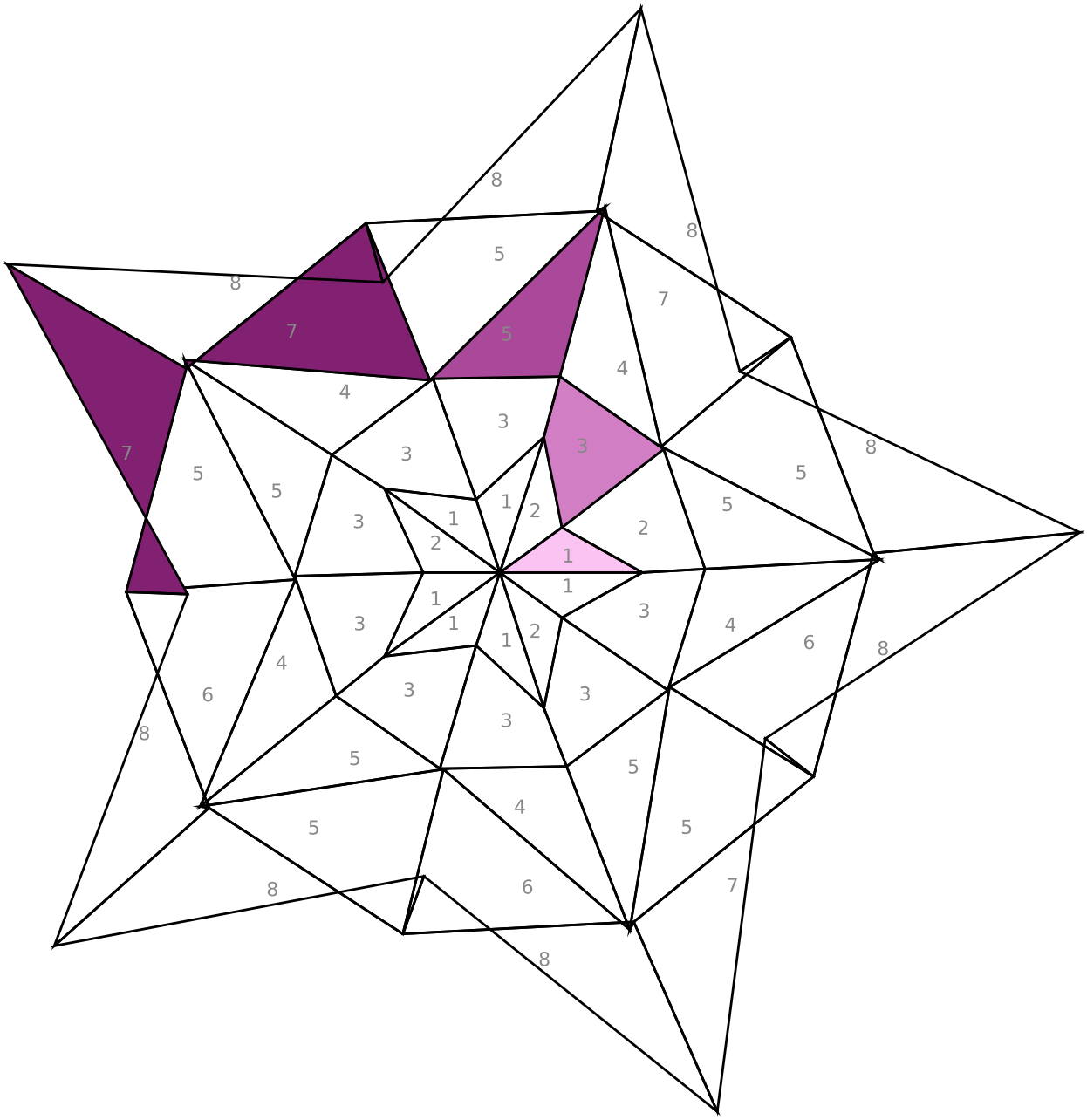
6. Decagram Star 1

1	2	3	4	5	6	7	8



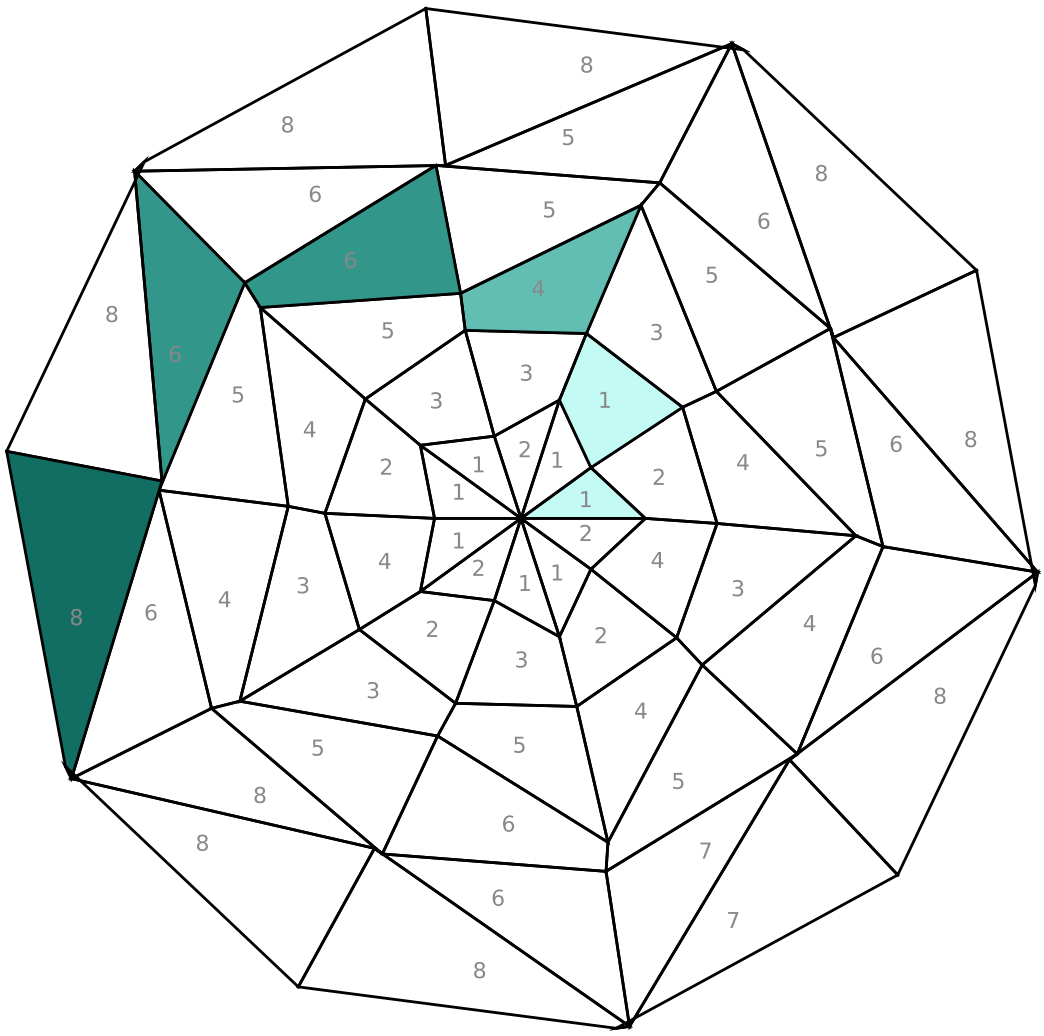
7. Decagram Star 2

1	2	3	4	5	6	7	8



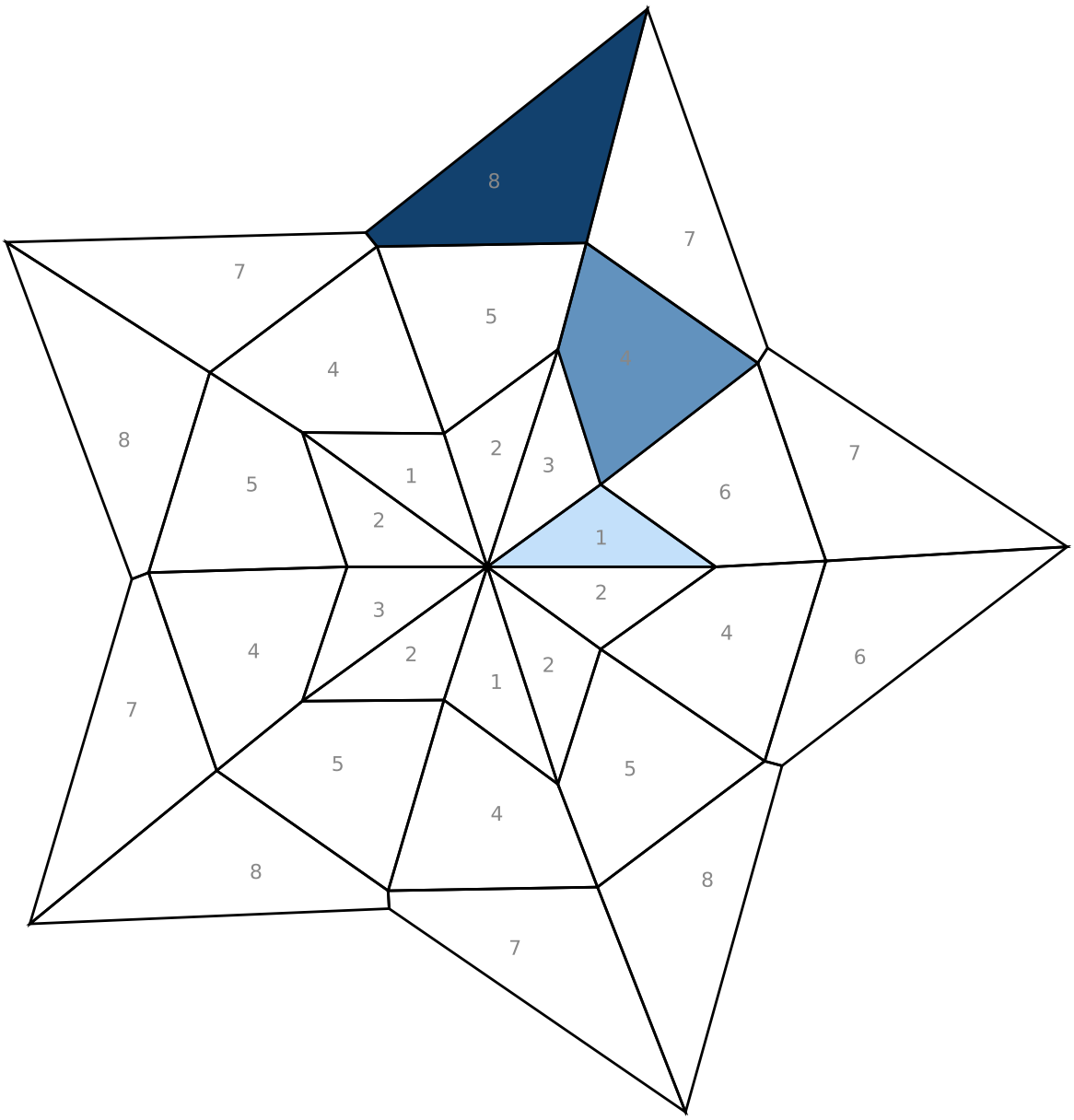
8. Decagram Star 3

1	2	3	4	5	6	7	8



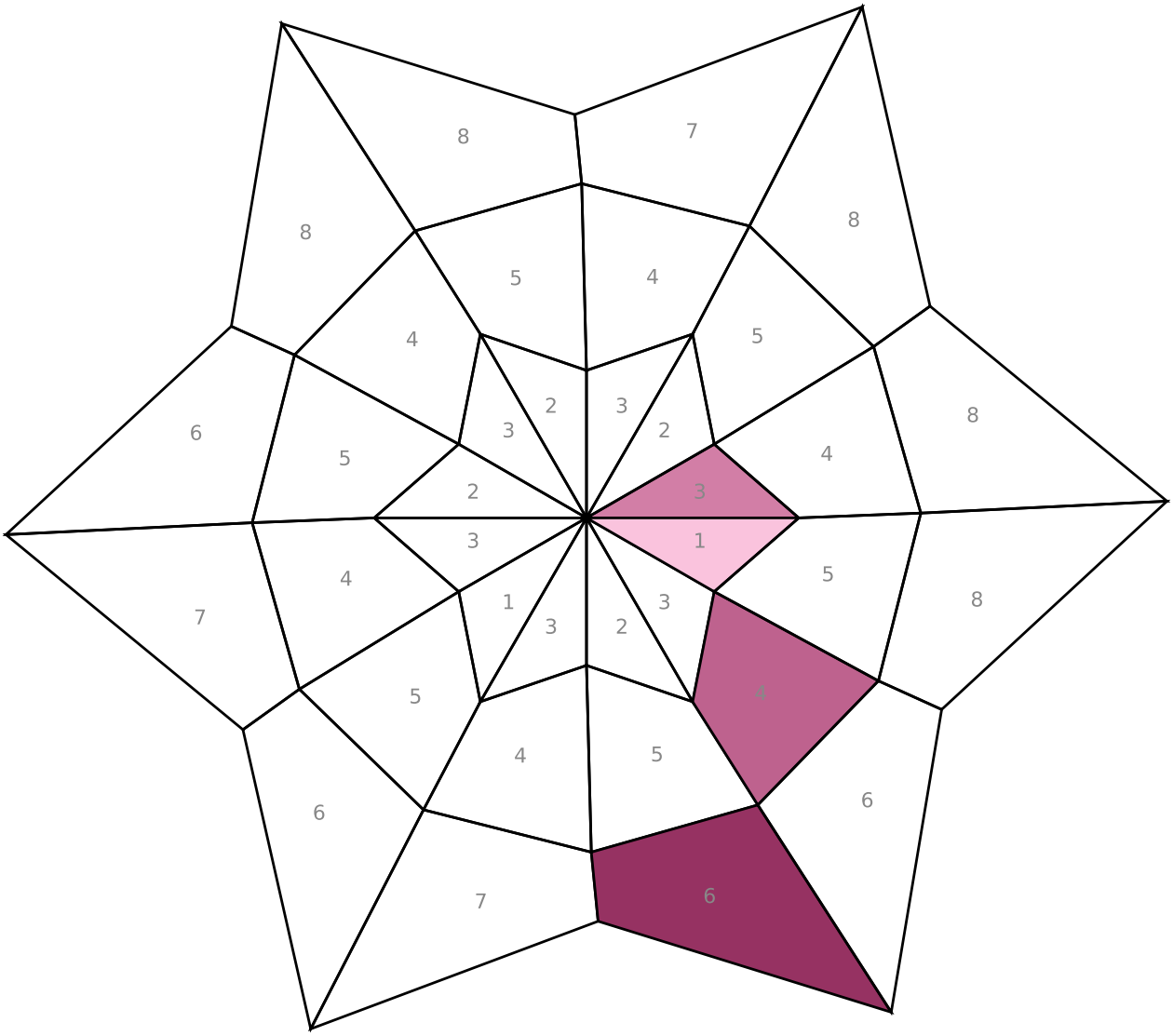
9. Decagram Star 4

1	2	3	4	5	6	7	8



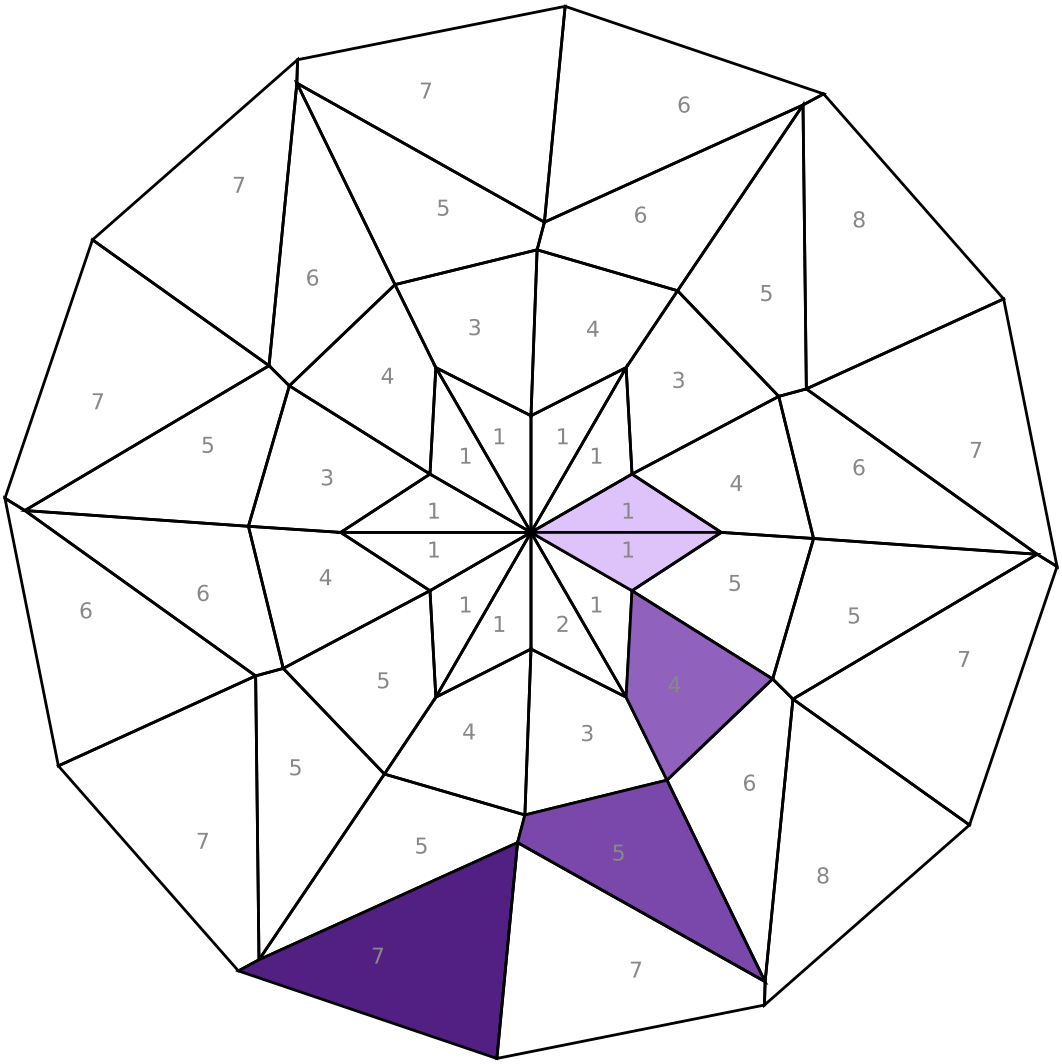
10. Decagram Star 5

1	2	3	4	5	6	7	8



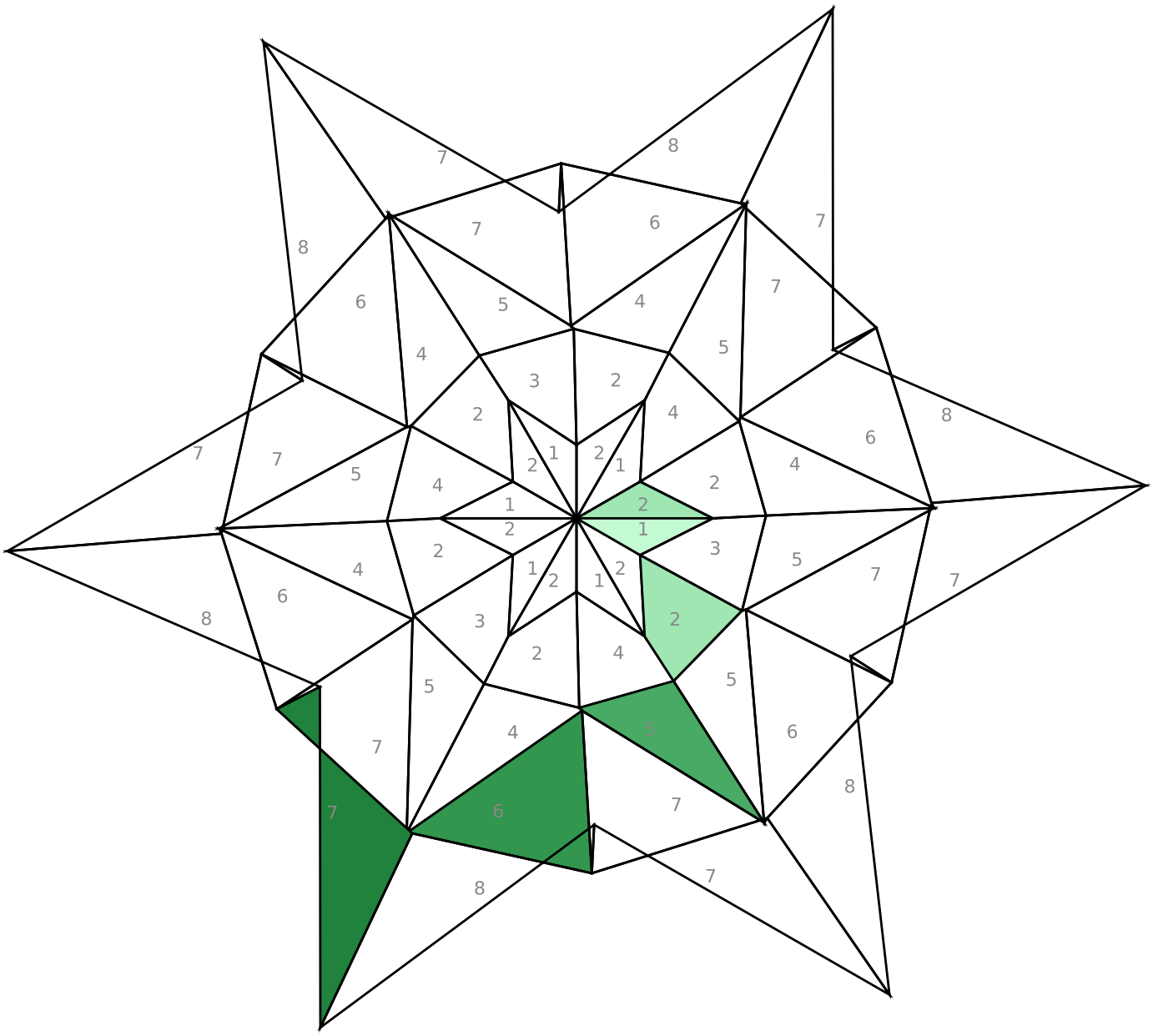
11. Dodecagram Mandala 1

1	2	3	4	5	6	7	8



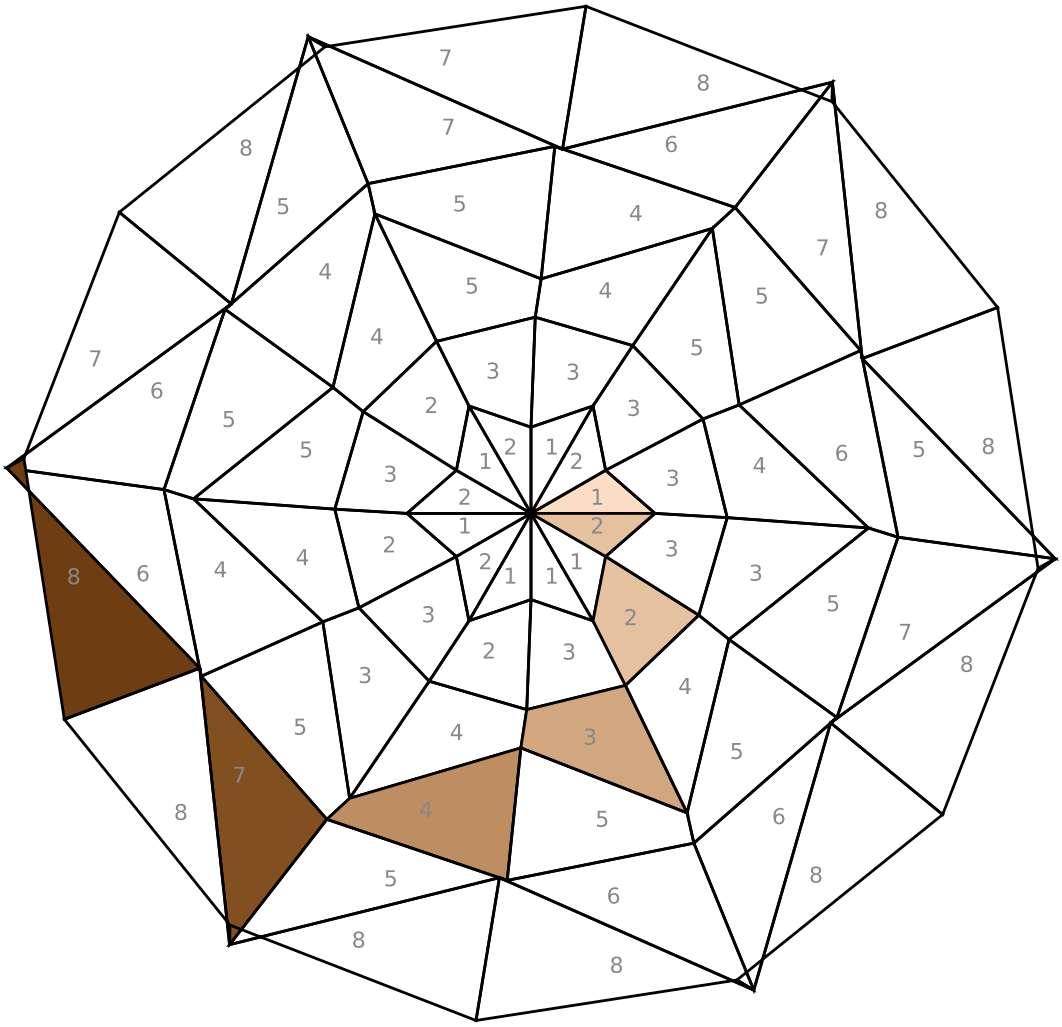
12. Dodecagram Mandala 2

1	2	3	4	5	6	7	8



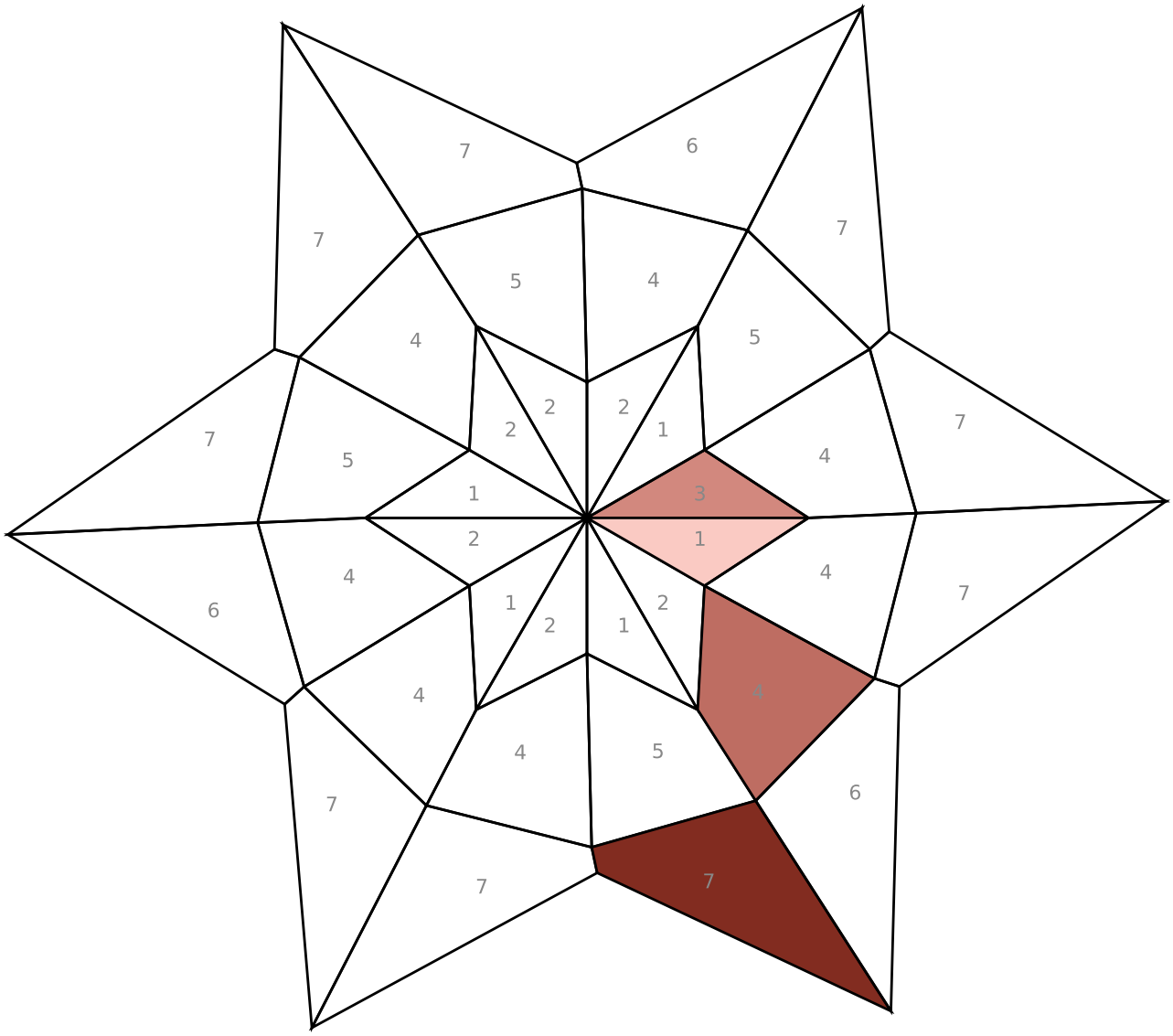
13. Dodecagram Mandala 3

1	2	3	4	5	6	7	8



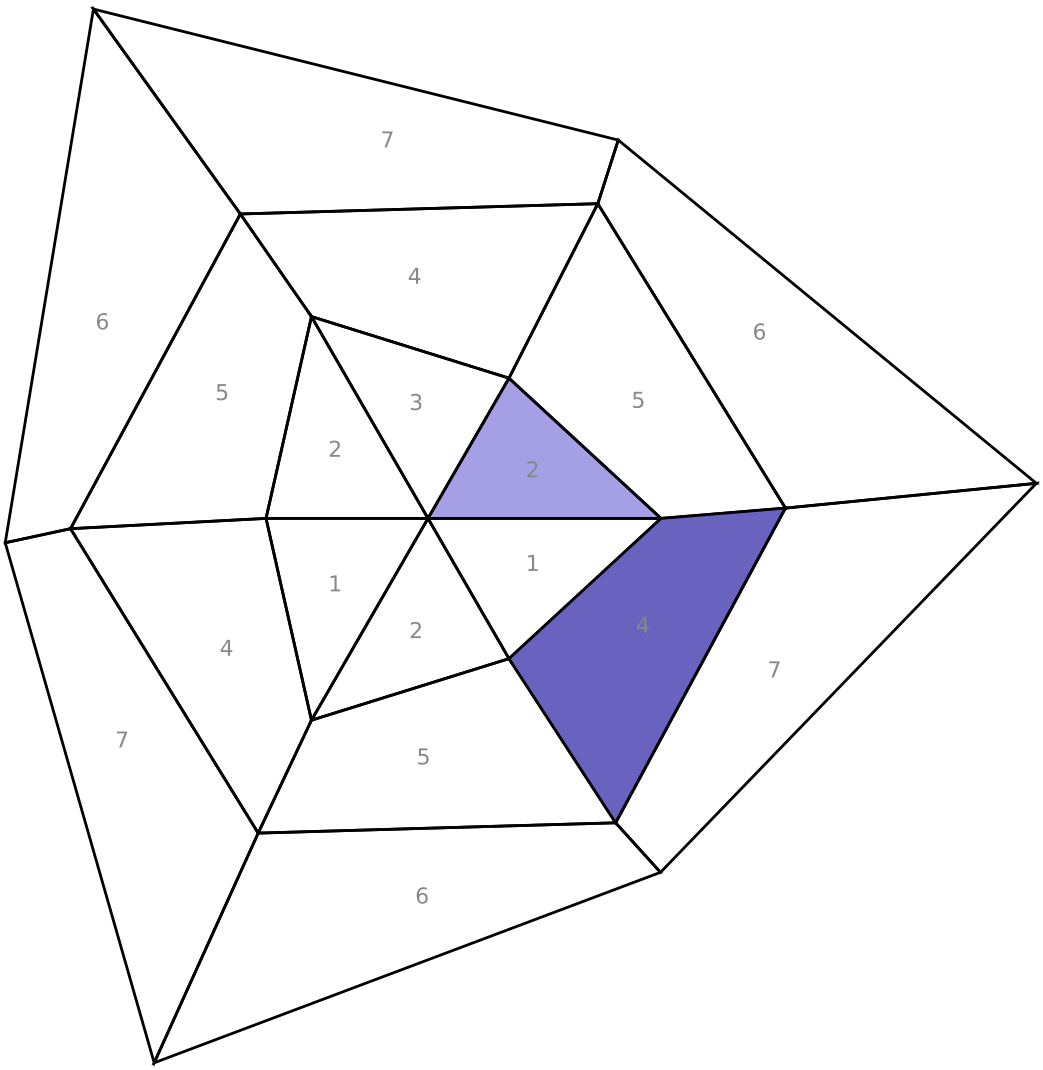
14. Dodecagram Mandala 4

1	2	3	4	5	6	7	8



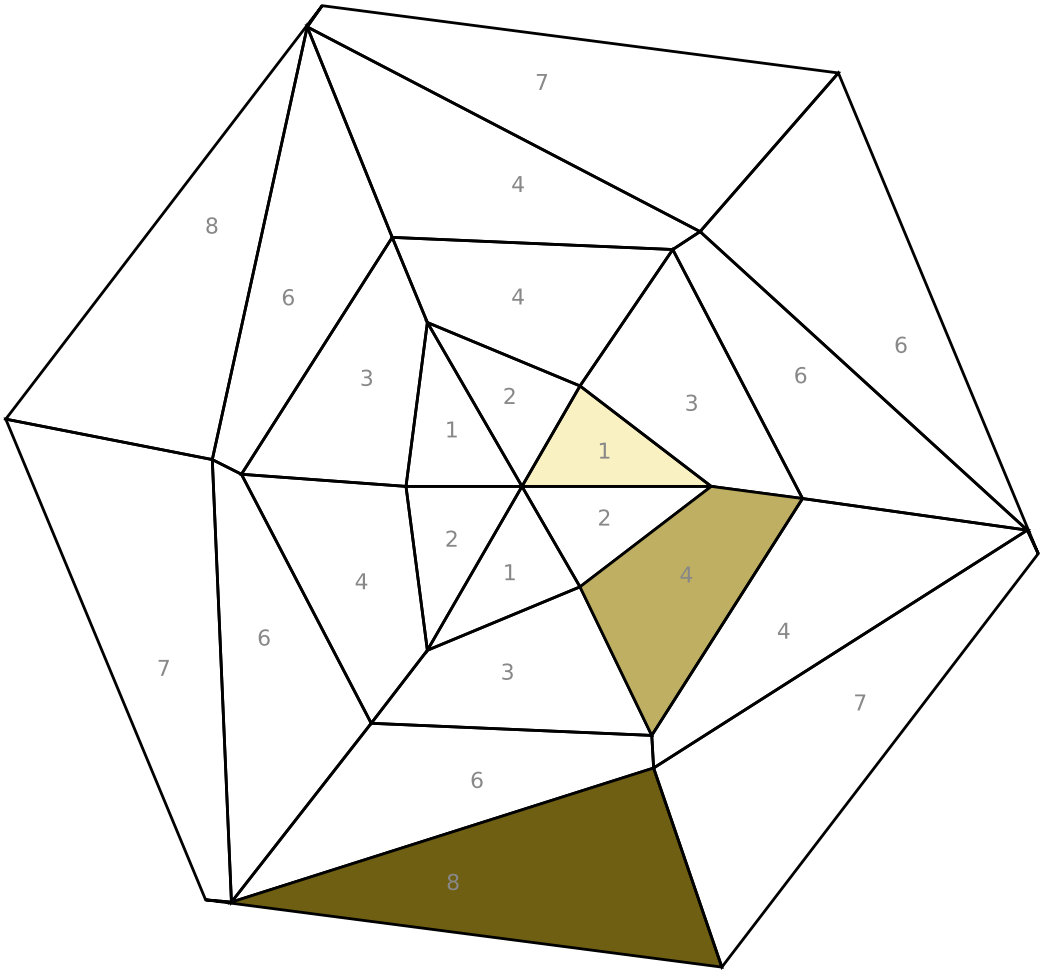
15. Dodecagram Mandala 5

1	2	3	4	5	6	7	8



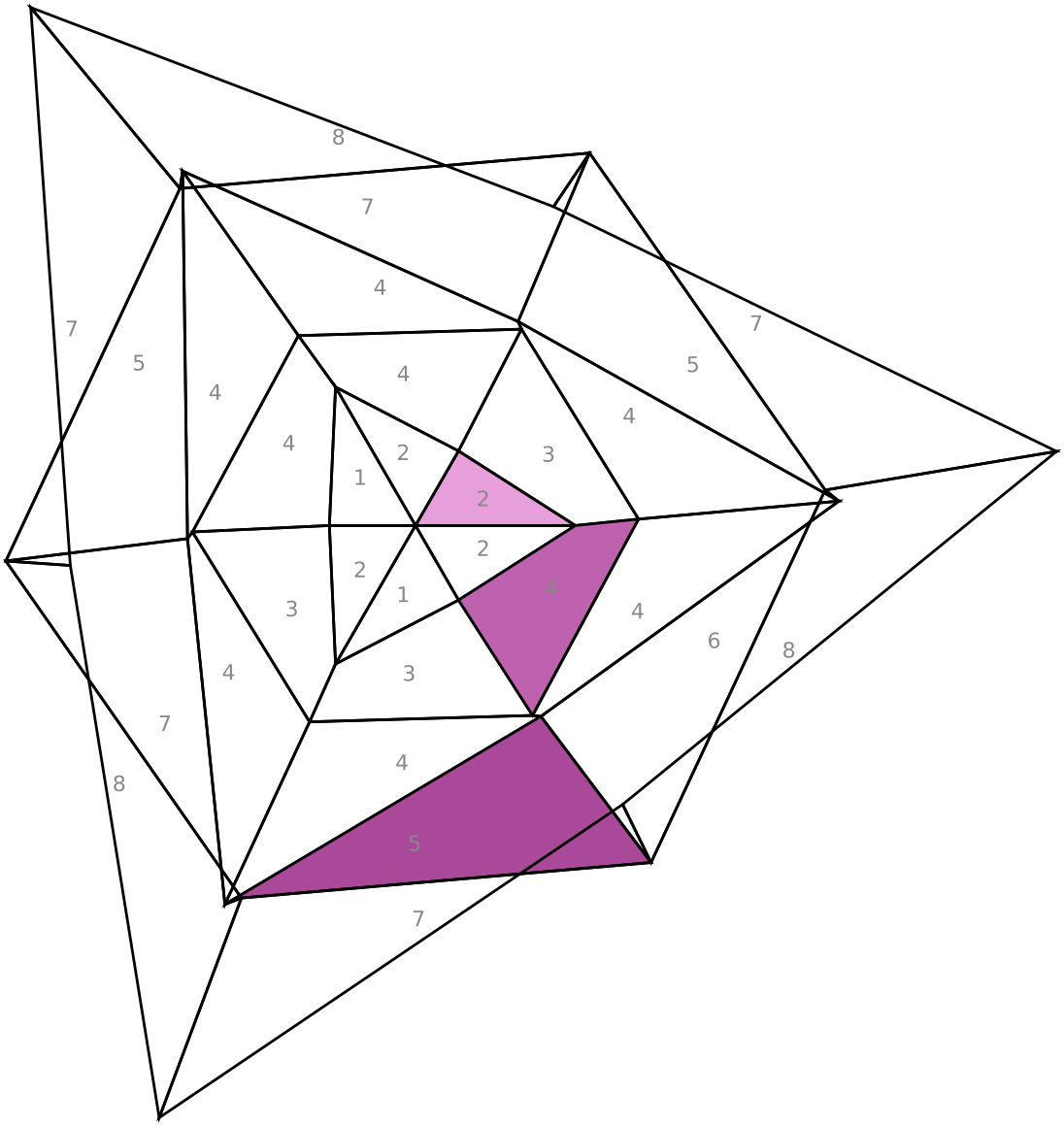
16. Hexagram Lattice 1

1	2	3	4	5	6	7	8



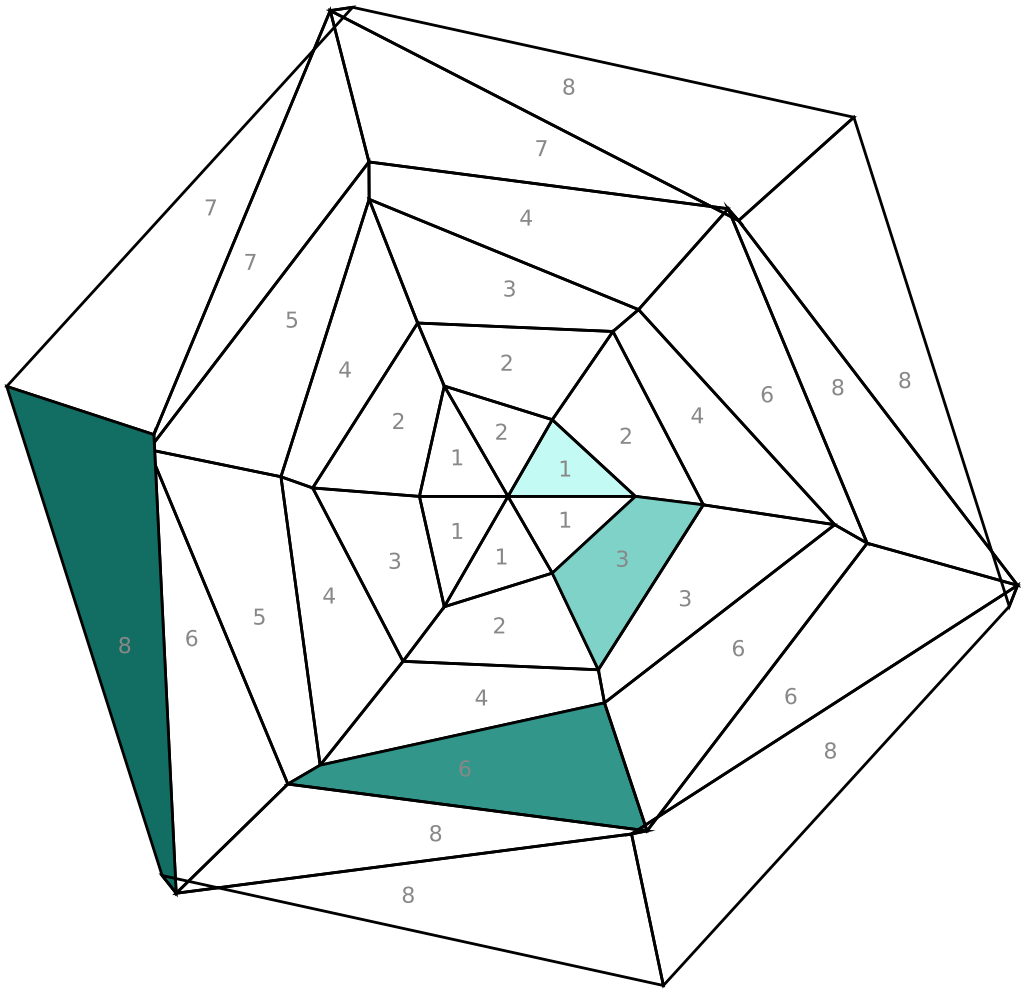
17. Hexagram Lattice 2

1	2	3	4	5	6	7	8



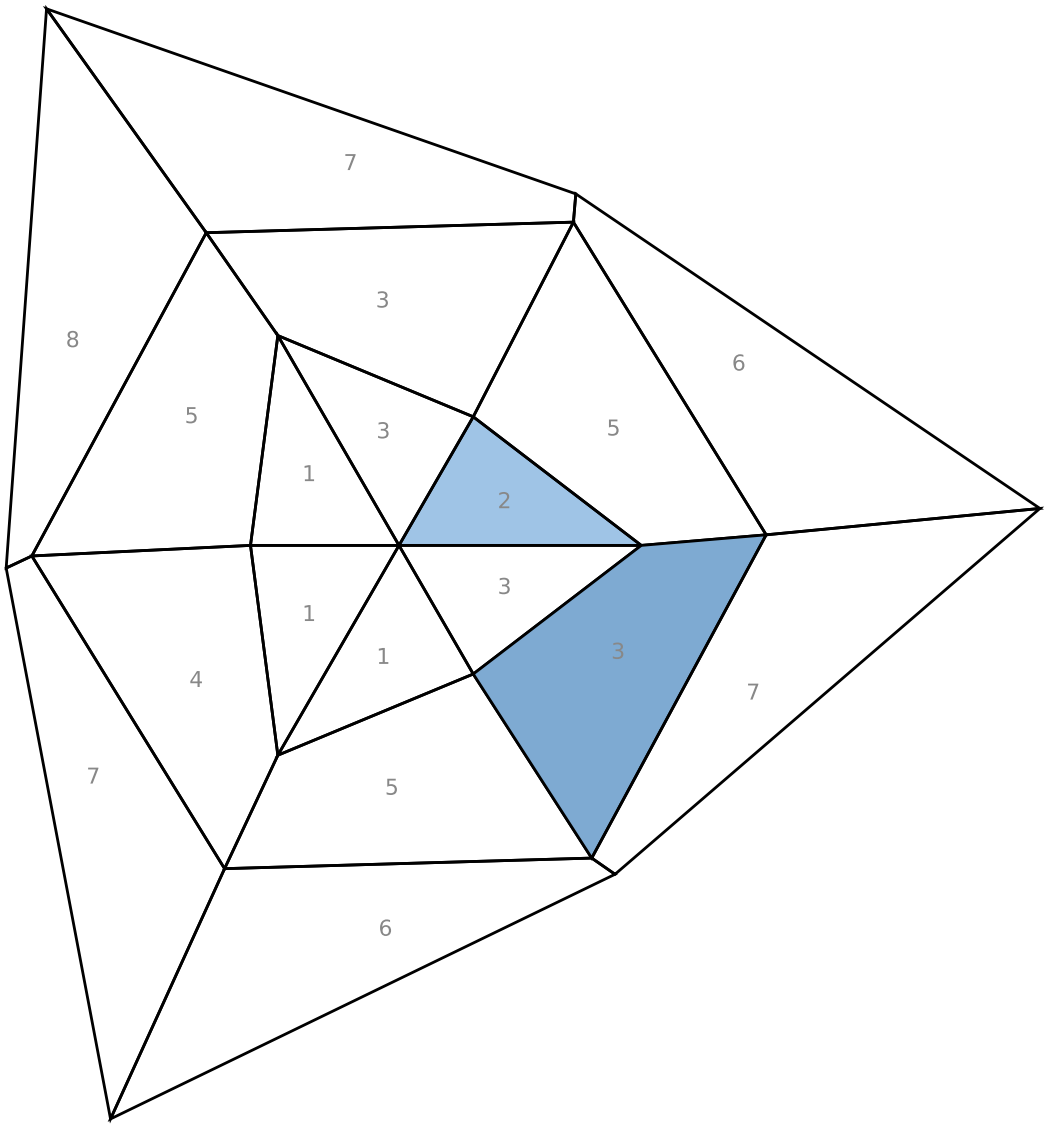
18. Hexagram Lattice 3

1	2	3	4	5	6	7	8



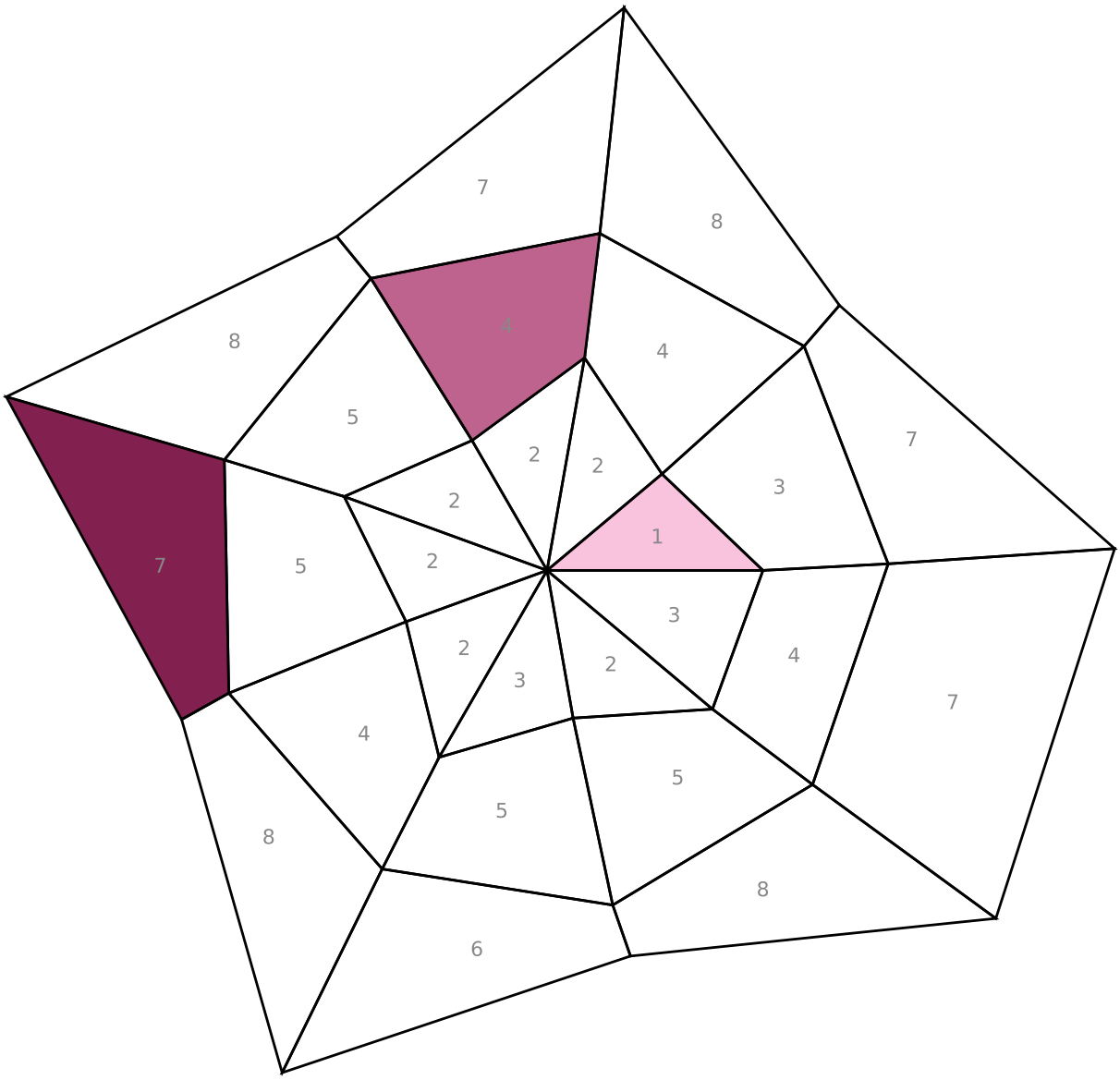
19. Hexagram Lattice 4

1	2	3	4	5	6	7	8



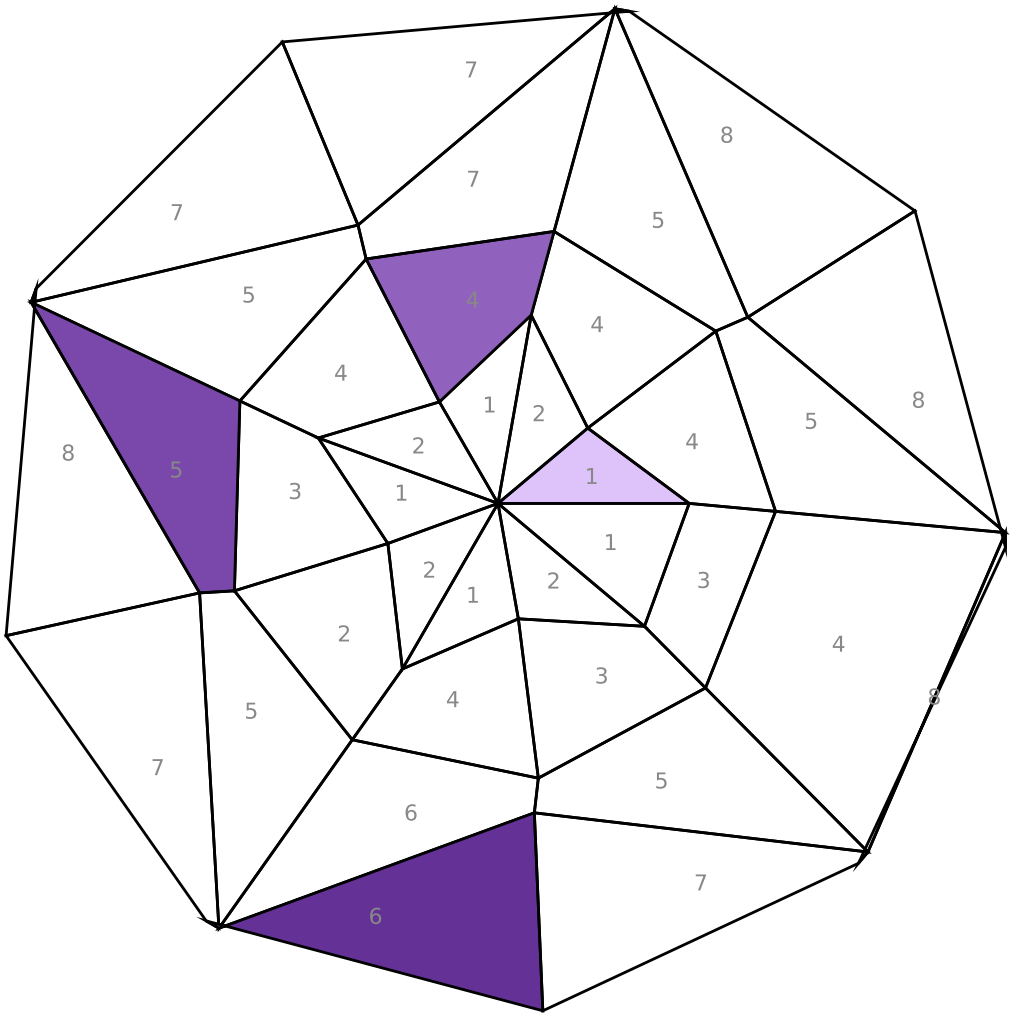
20. Hexagram Lattice 5

1	2	3	4	5	6	7	8



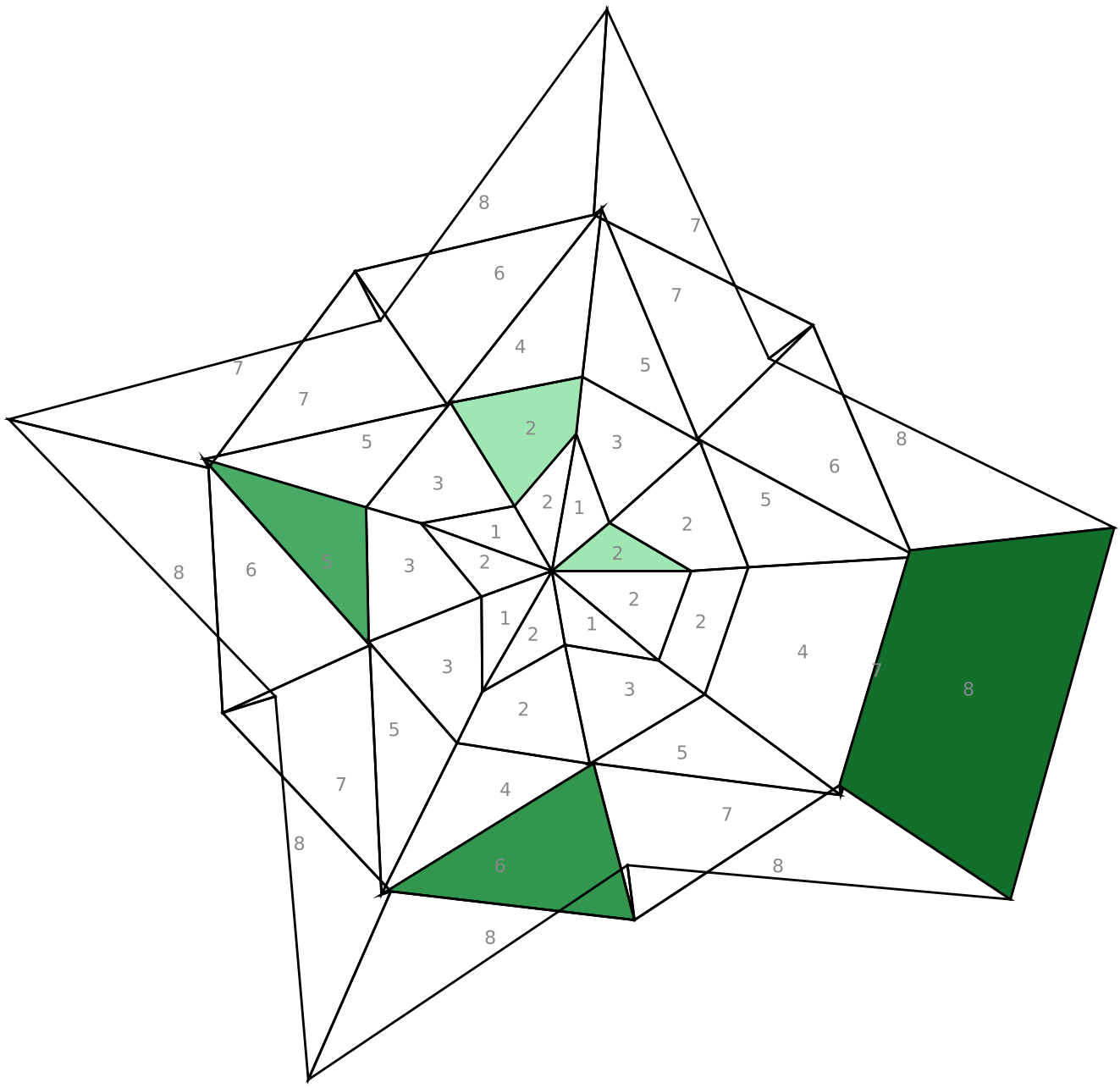
21. Enneagram Weave 1

1	2	3	4	5	6	7	8



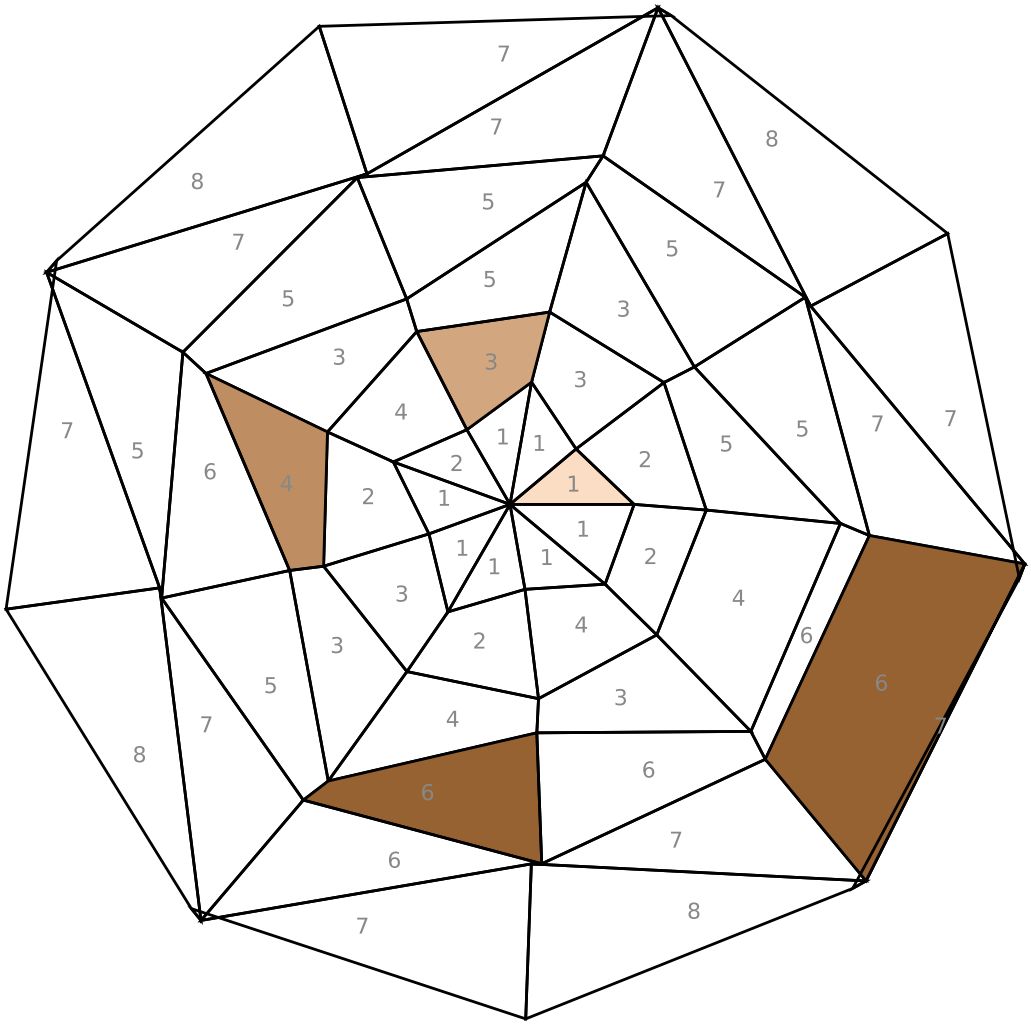
22. Enneagram Weave 2

1	2	3	4	5	6	7	8



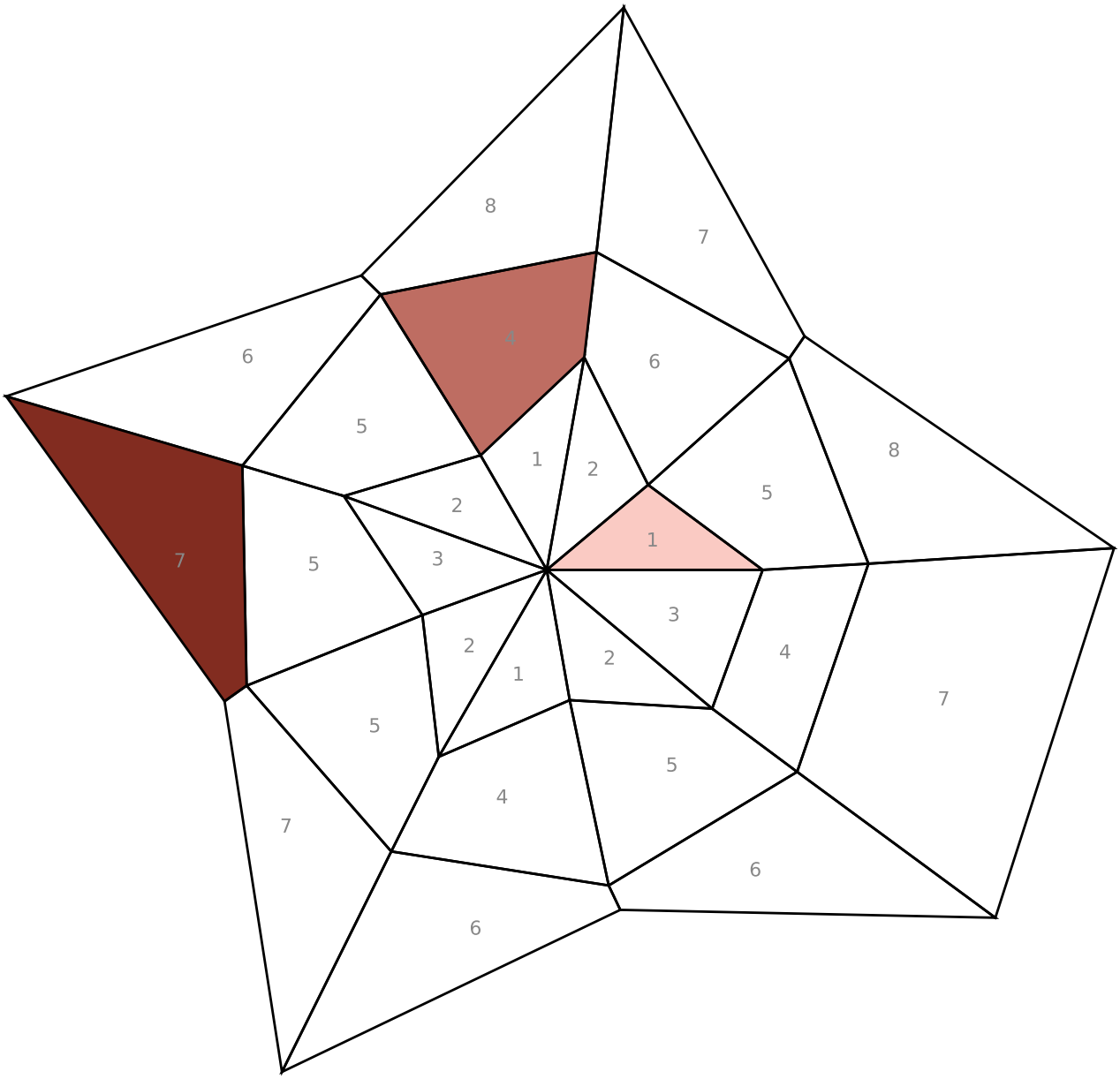
23. Enneagram Weave 3

1	2	3	4	5	6	7	8



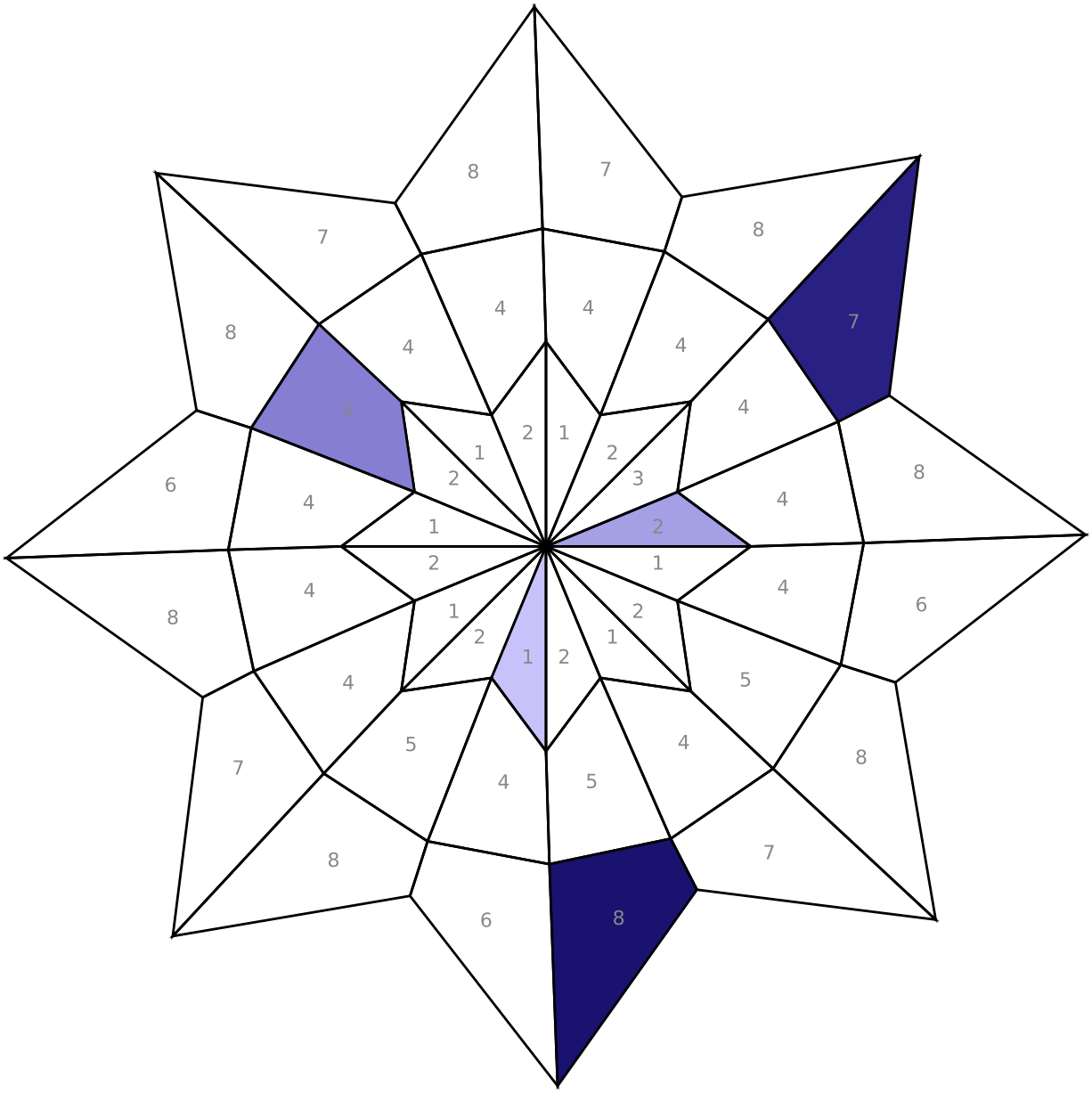
24. Enneagram Weave 4

1	2	3	4	5	6	7	8



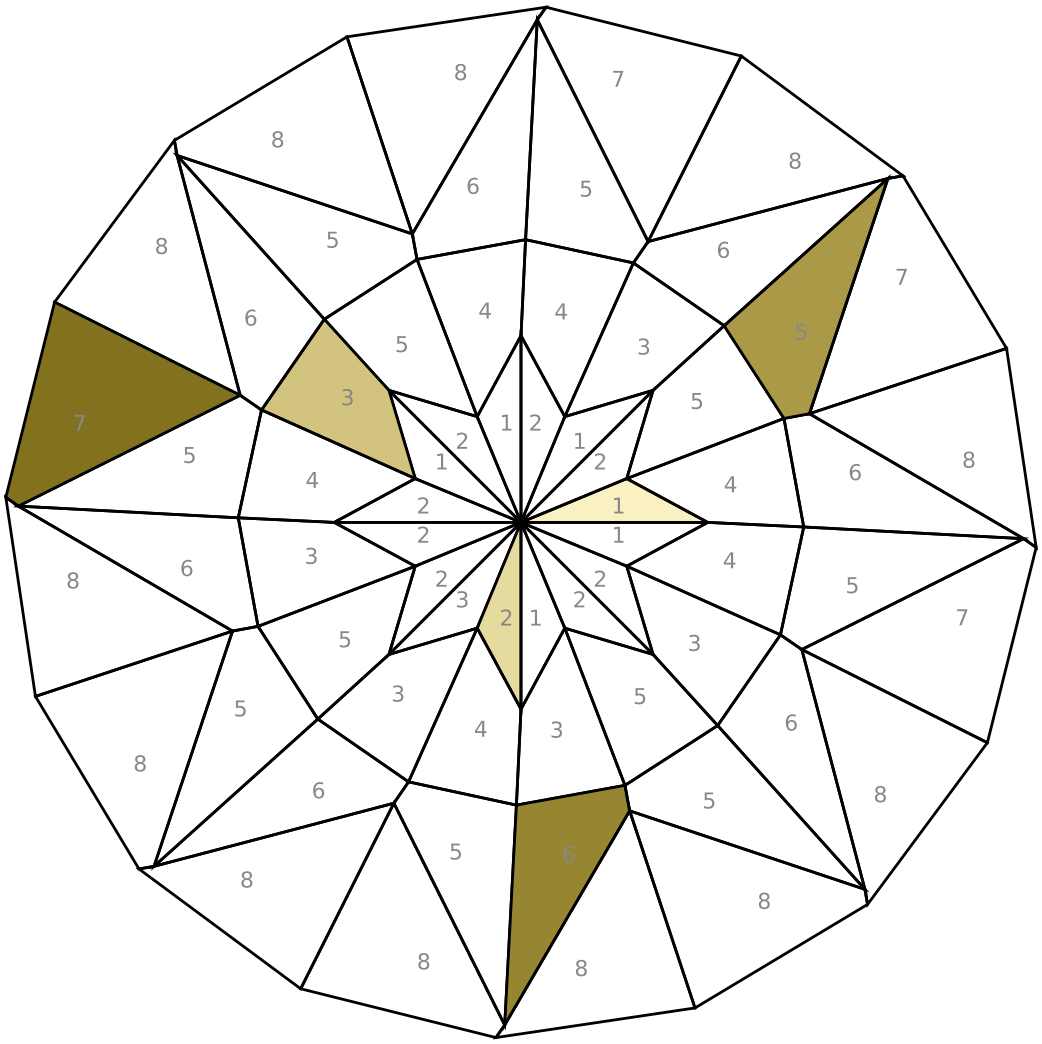
25. Enneagram Weave 5

1	2	3	4	5	6	7	8



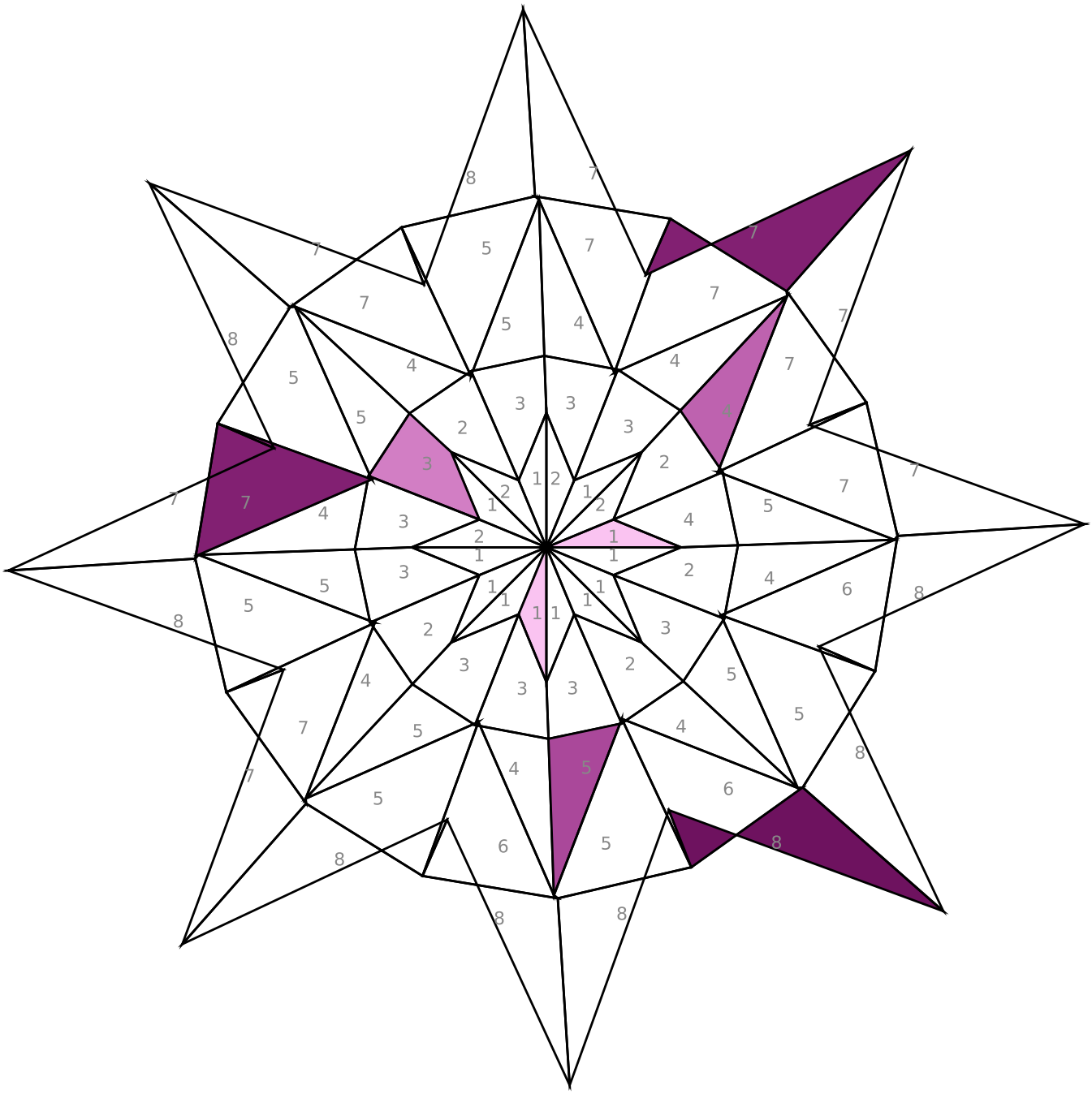
26. Sixteen-Point Star 1

1	2	3	4	5	6	7	8



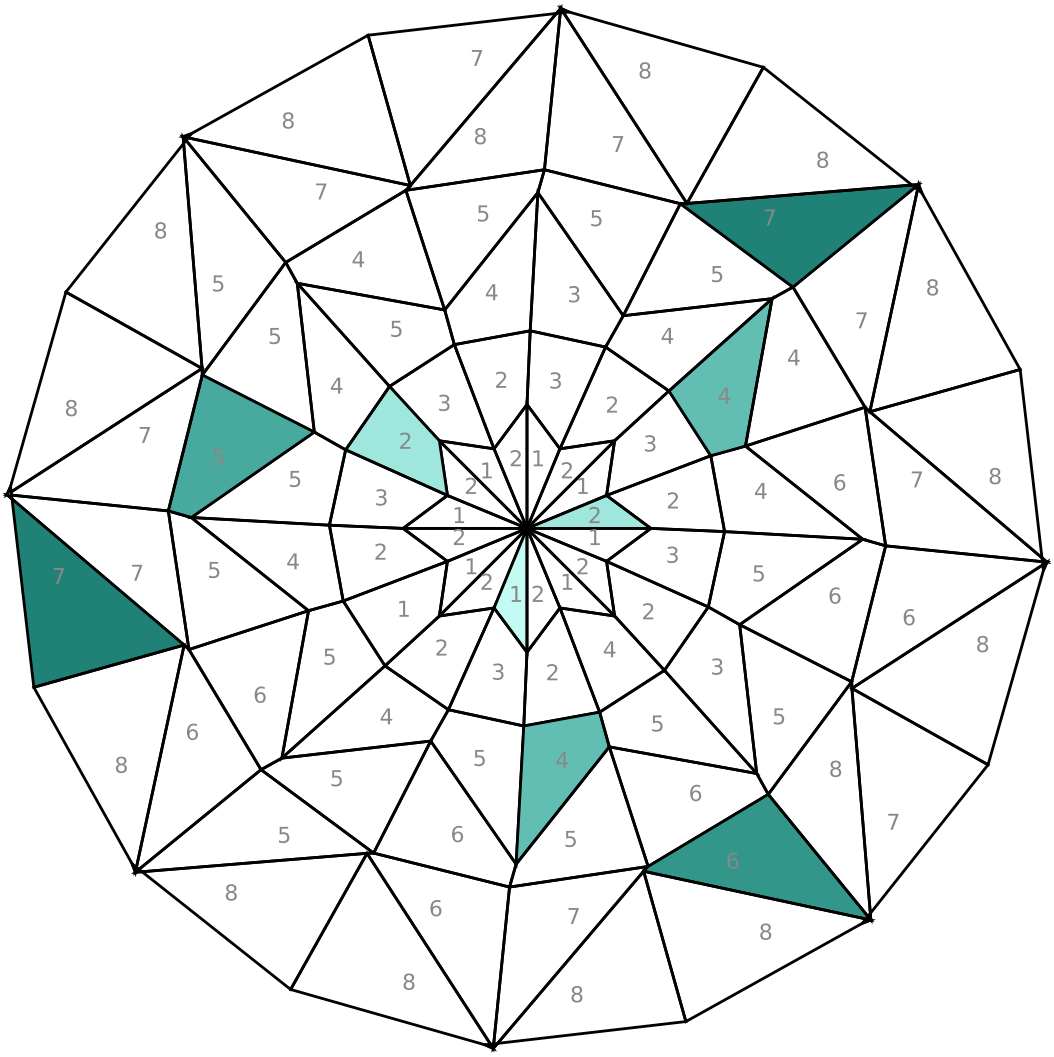
27. Sixteen-Point Star 2

1	2	3	4	5	6	7	8



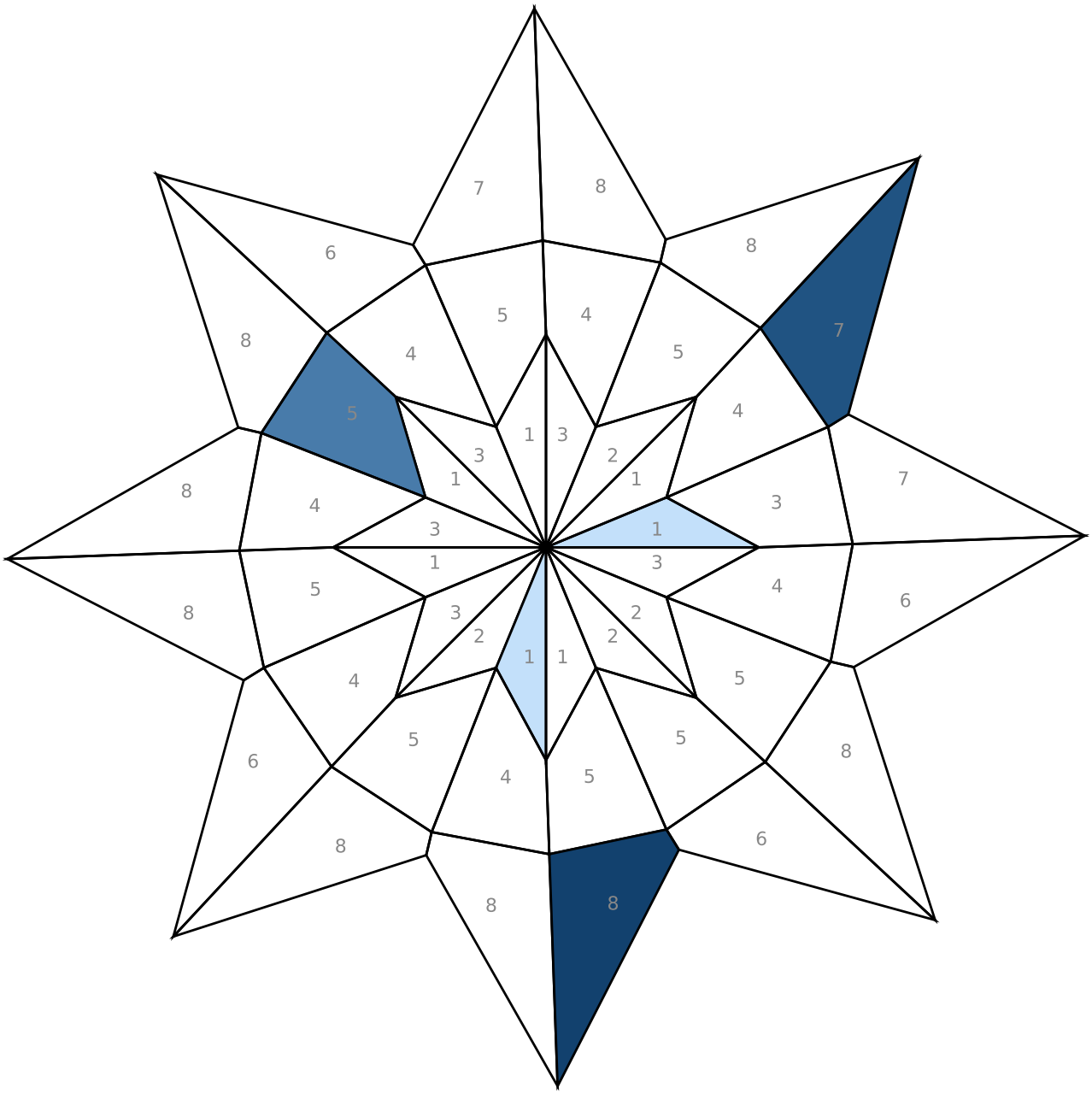
28. Sixteen-Point Star 3

1	2	3	4	5	6	7	8



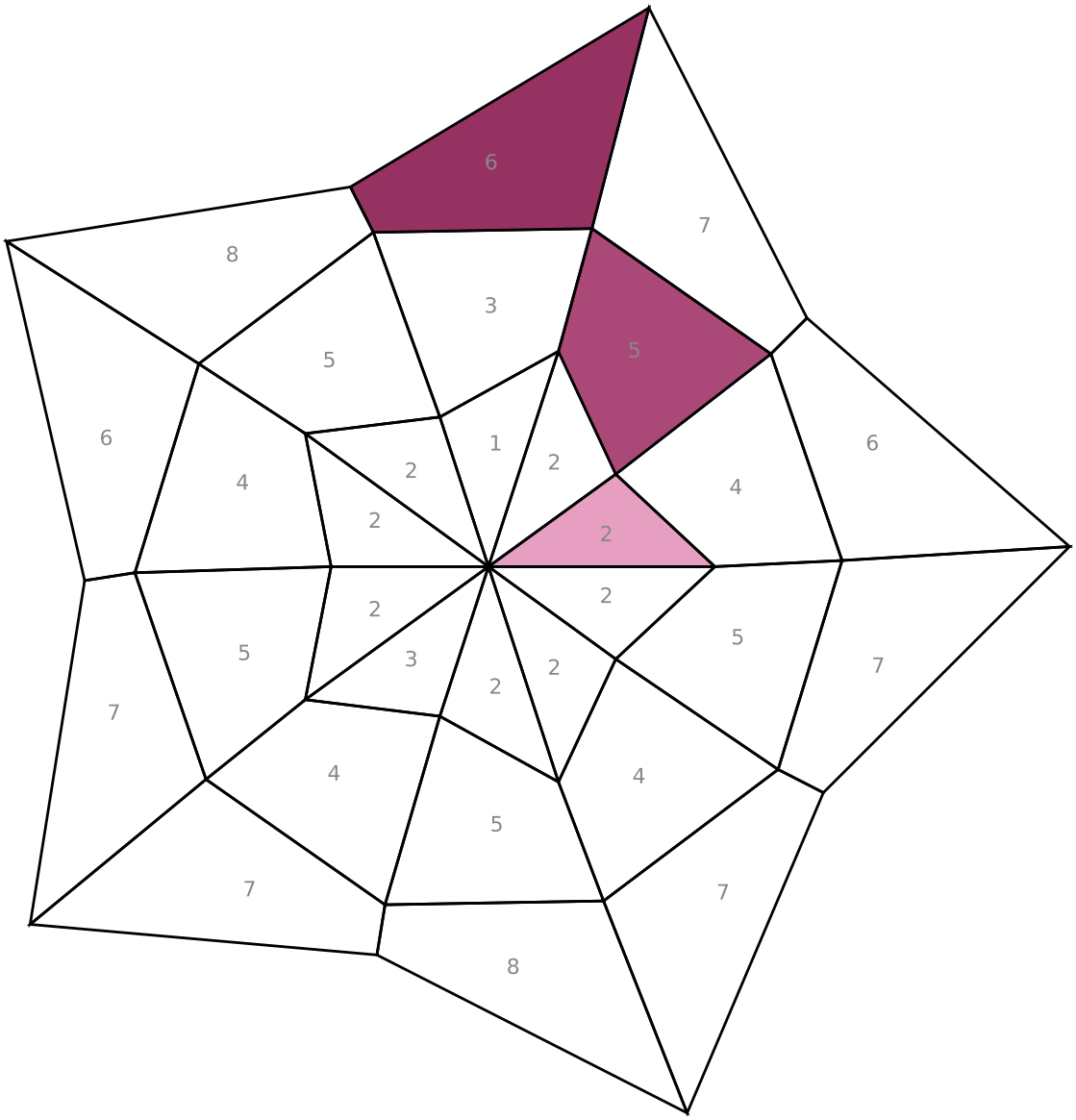
29. Sixteen-Point Star 4

1	2	3	4	5	6	7	8



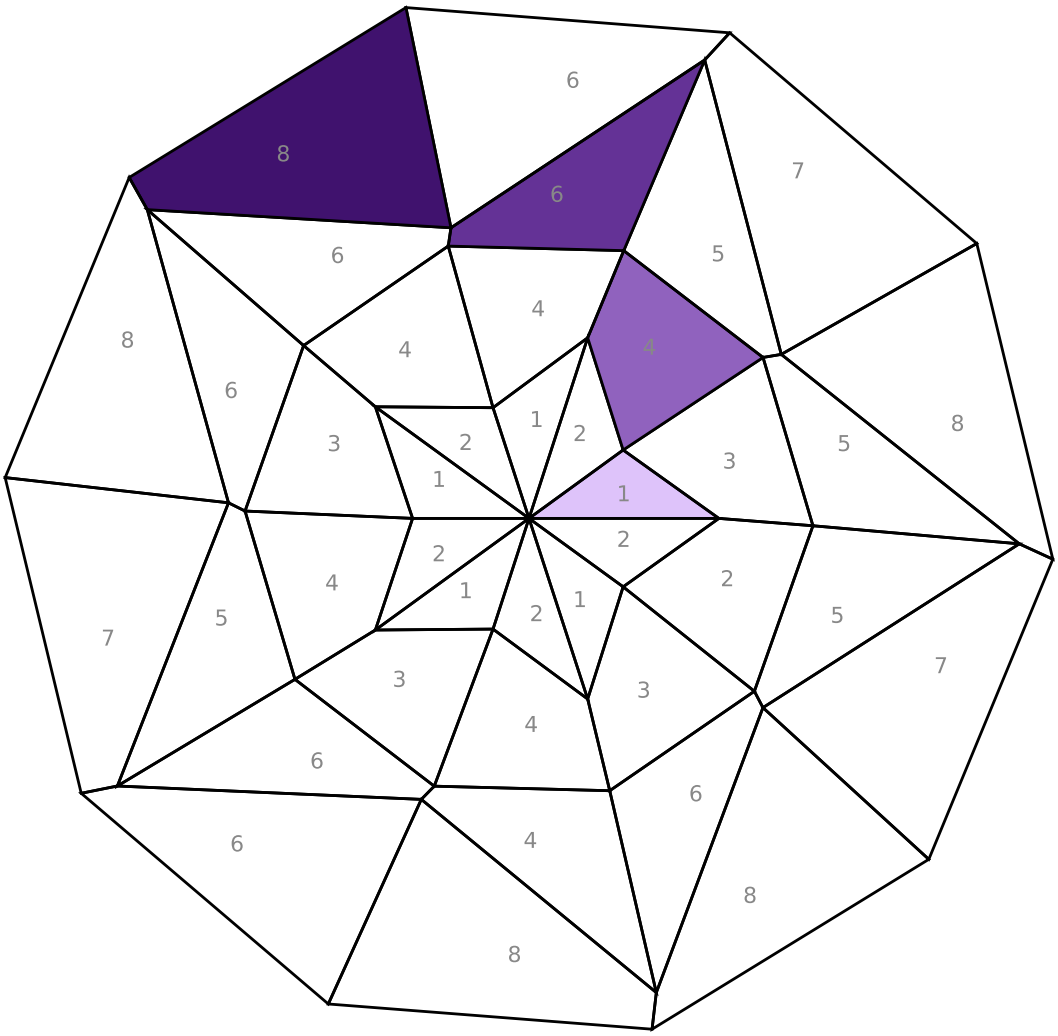
30. Sixteen-Point Star 5

1	2	3	4	5	6	7	8



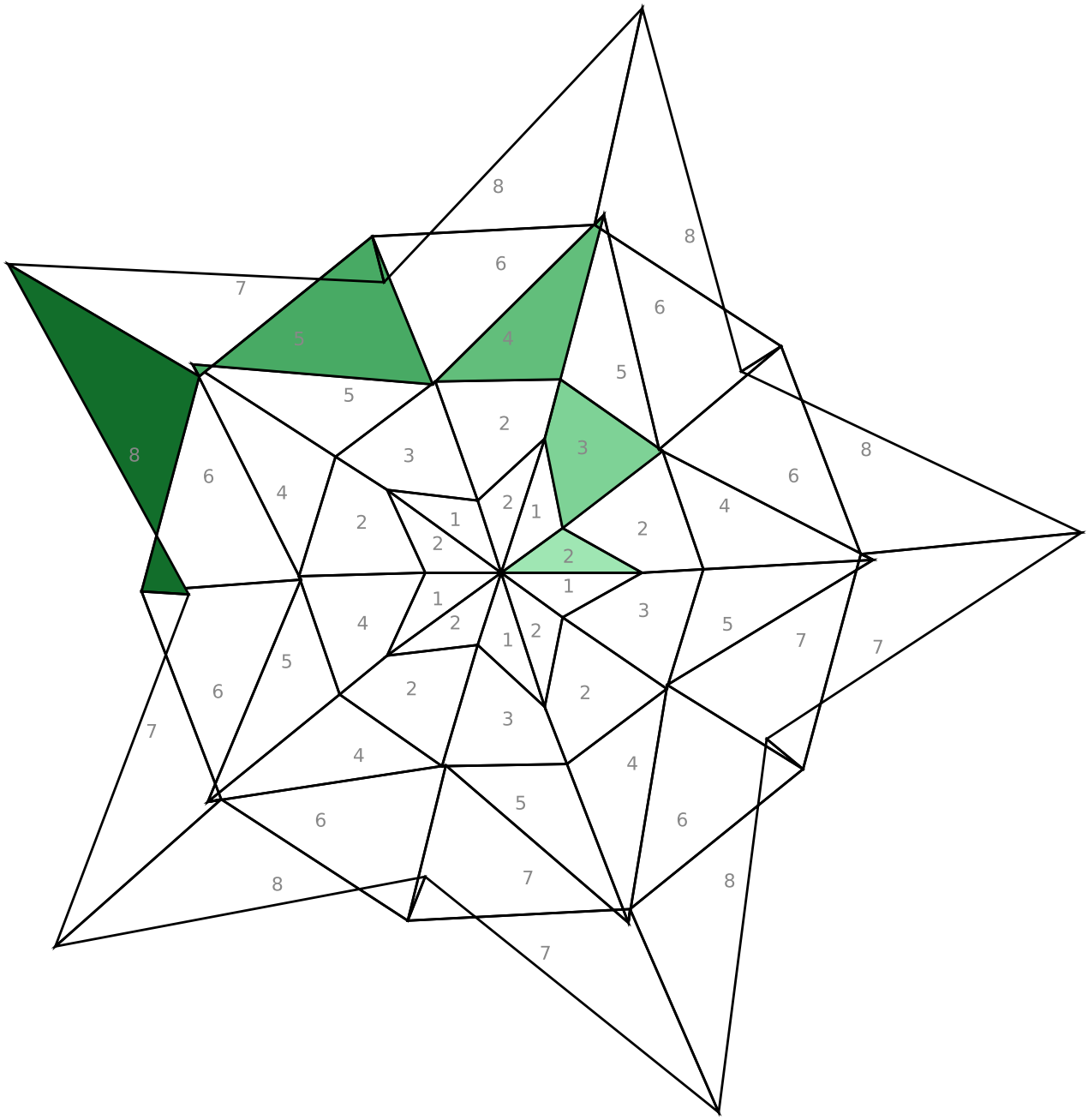
31. Girih Burst 1

1	2	3	4	5	6	7	8



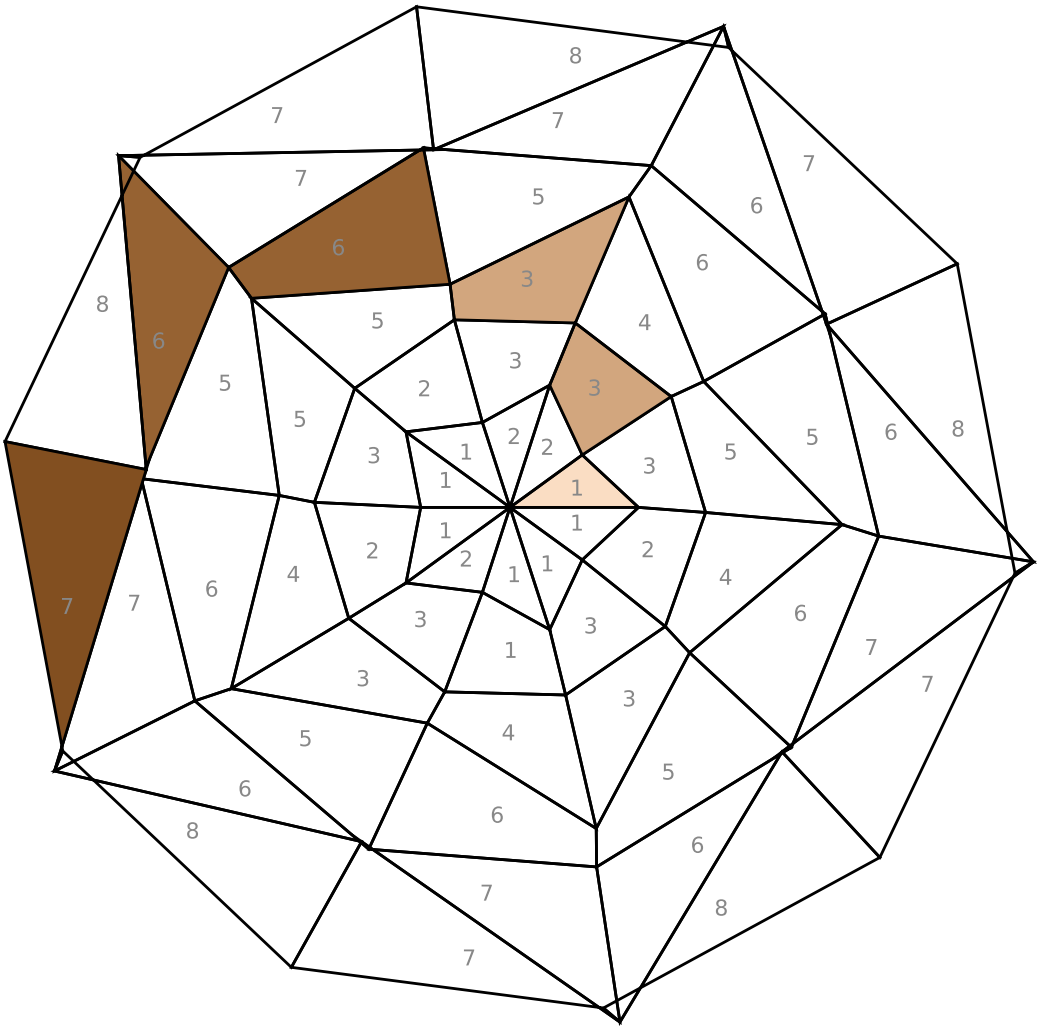
32. Girih Burst 2

1	2	3	4	5	6	7	8



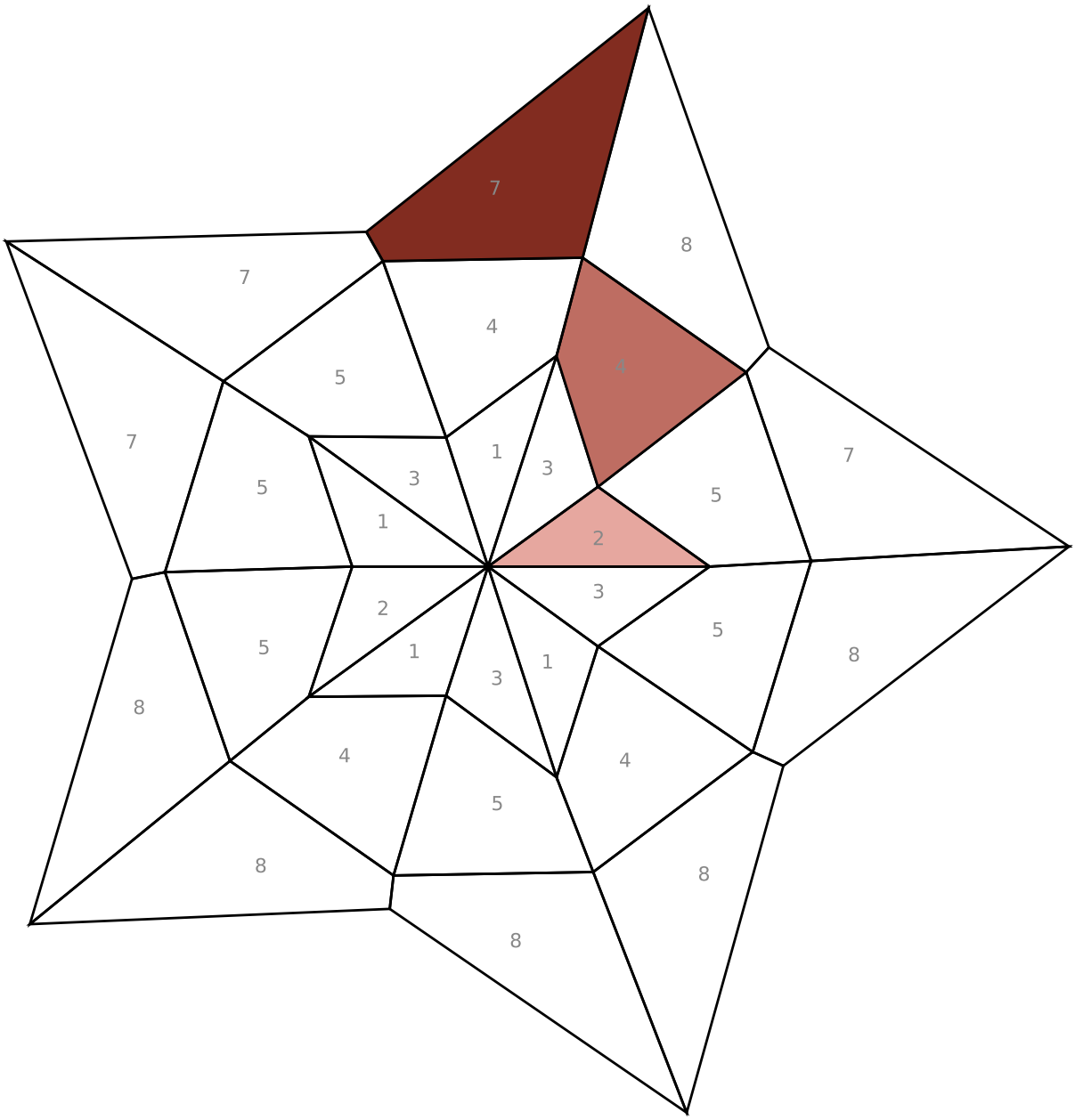
33. Girih Burst 3

1	2	3	4	5	6	7	8



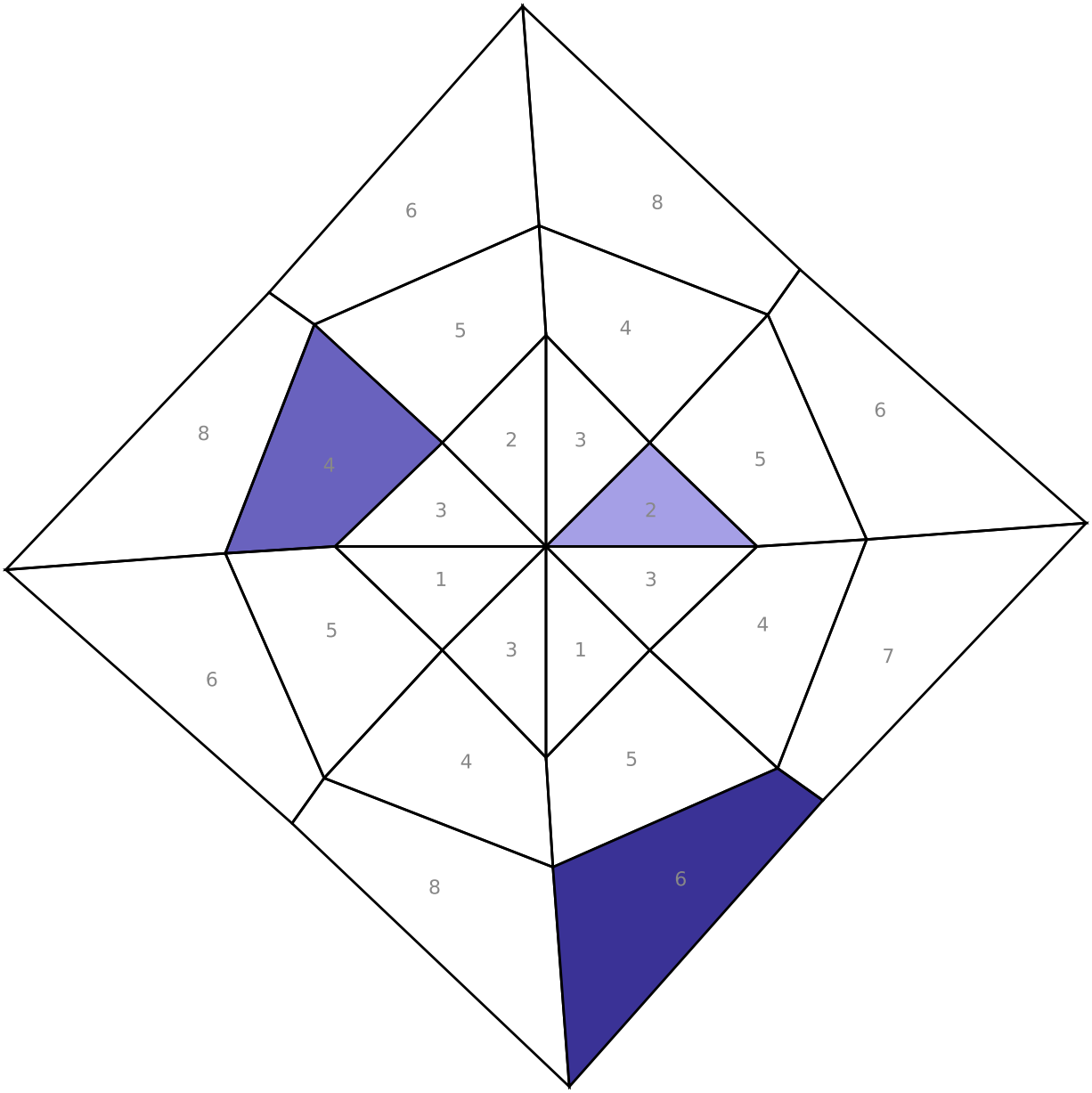
34. Girih Burst 4

1	2	3	4	5	6	7	8



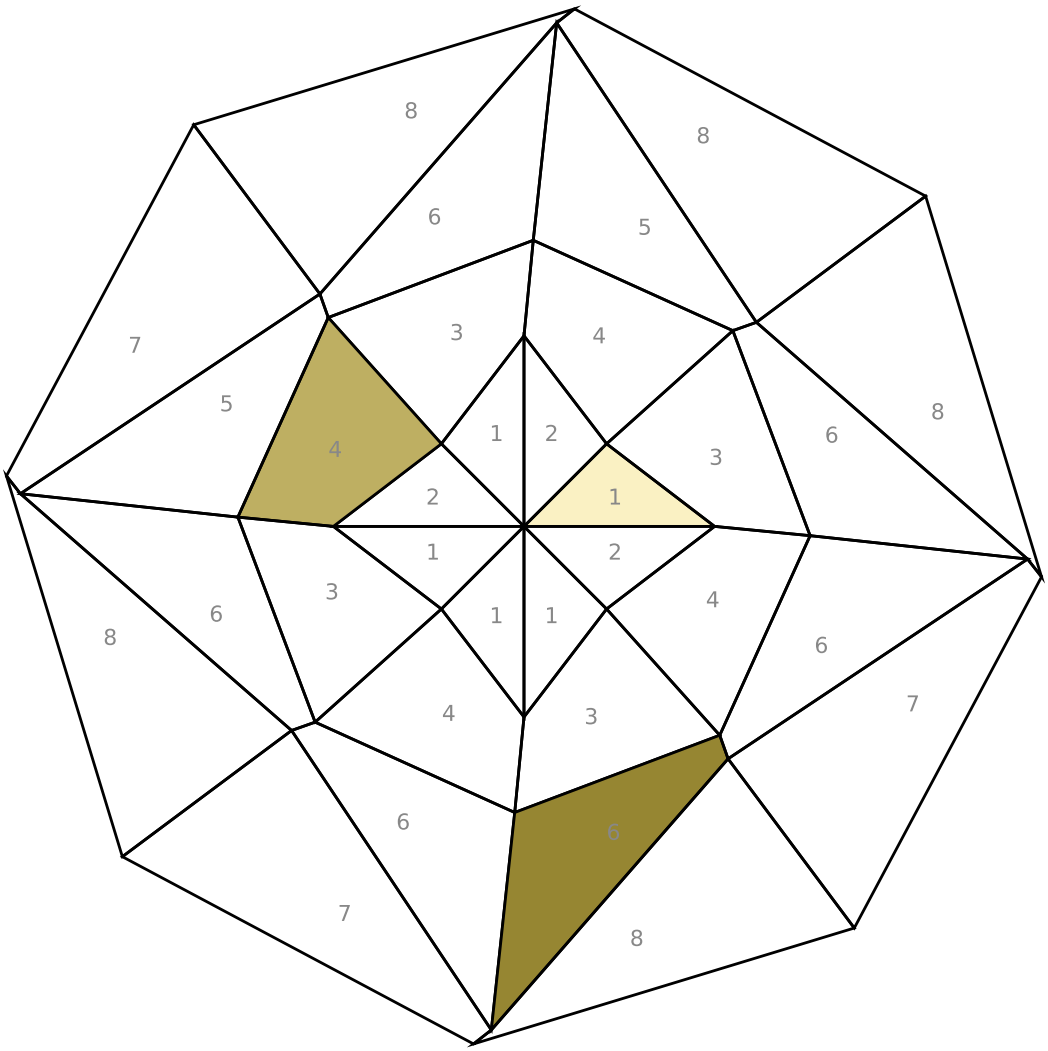
35. Girih Burst 5

1	2	3	4	5	6	7	8



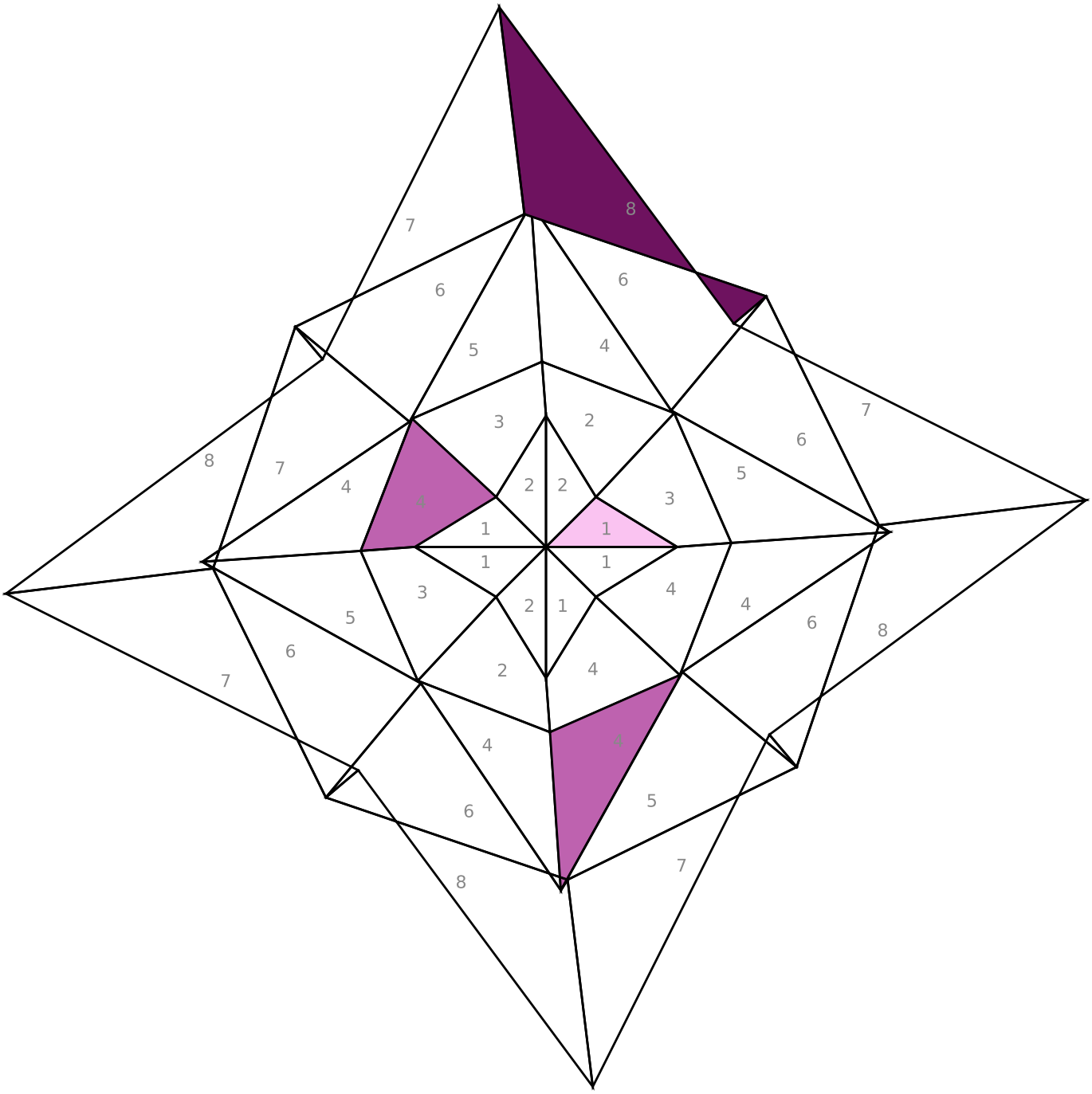
36. Rub el Hizb Star 1

1	2	3	4	5	6	7	8



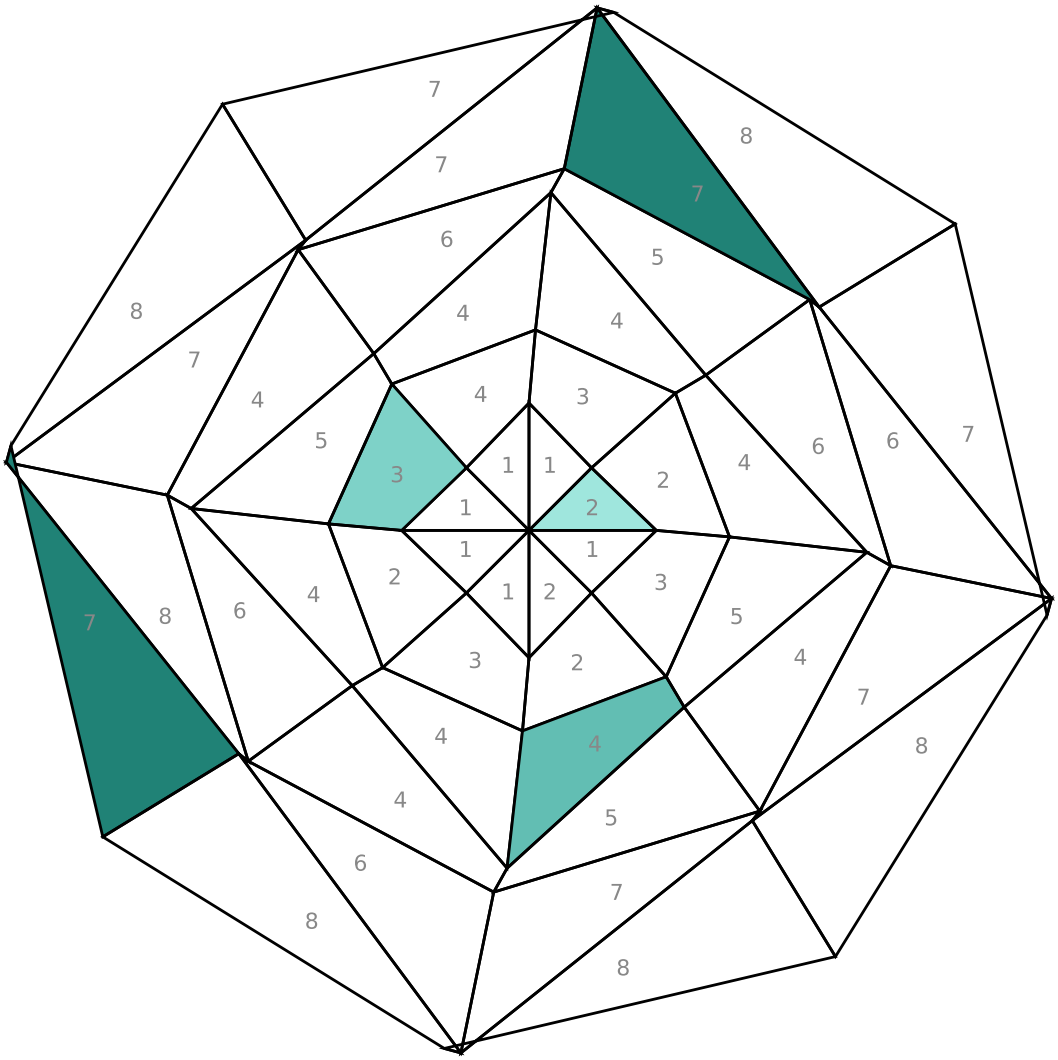
37. Rub el Hizb Star 2

1	2	3	4	5	6	7	8



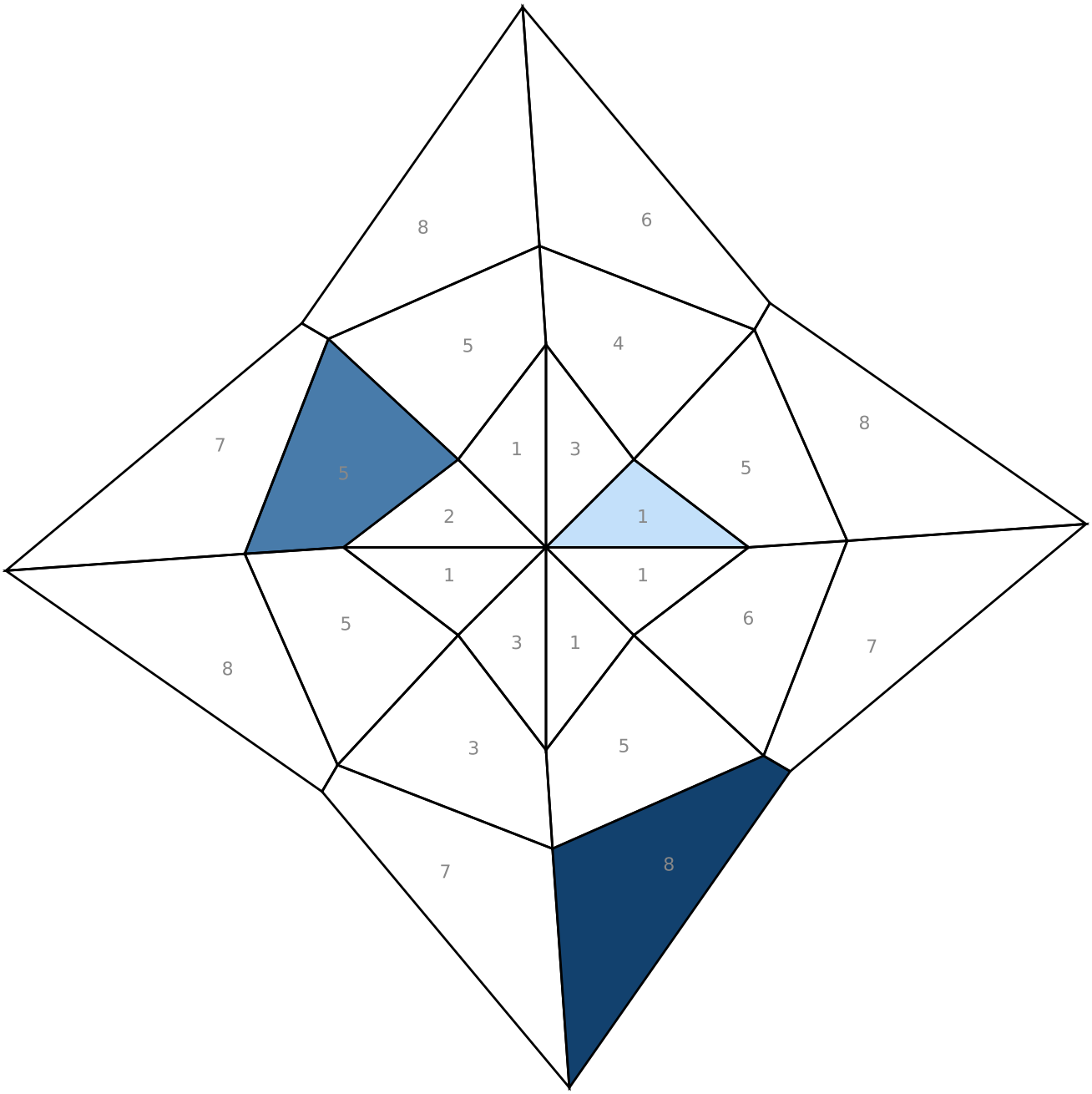
38. Rub el Hizb Star 3

1	2	3	4	5	6	7	8



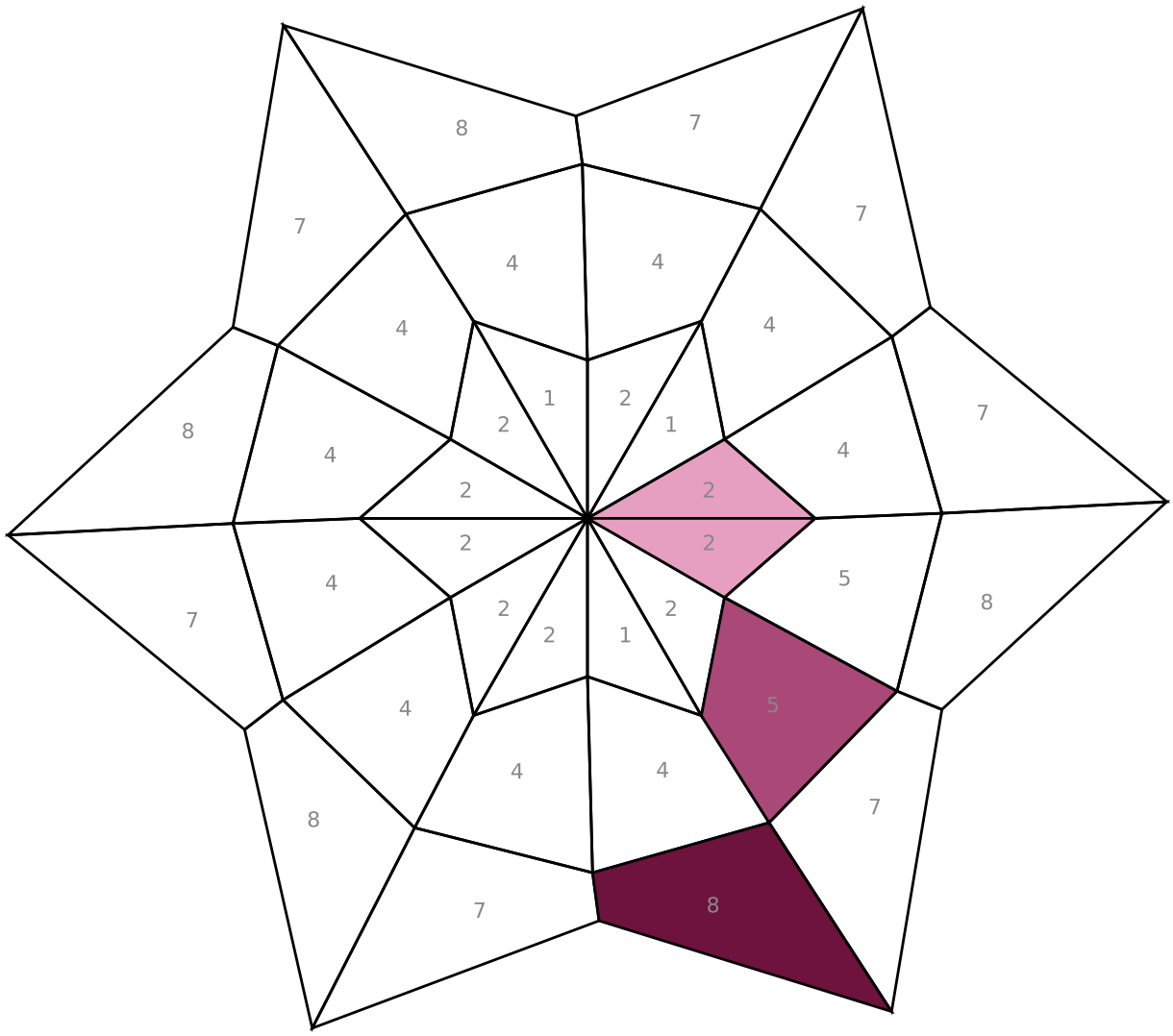
39. Rub el Hizb Star 4

1	2	3	4	5	6	7	8



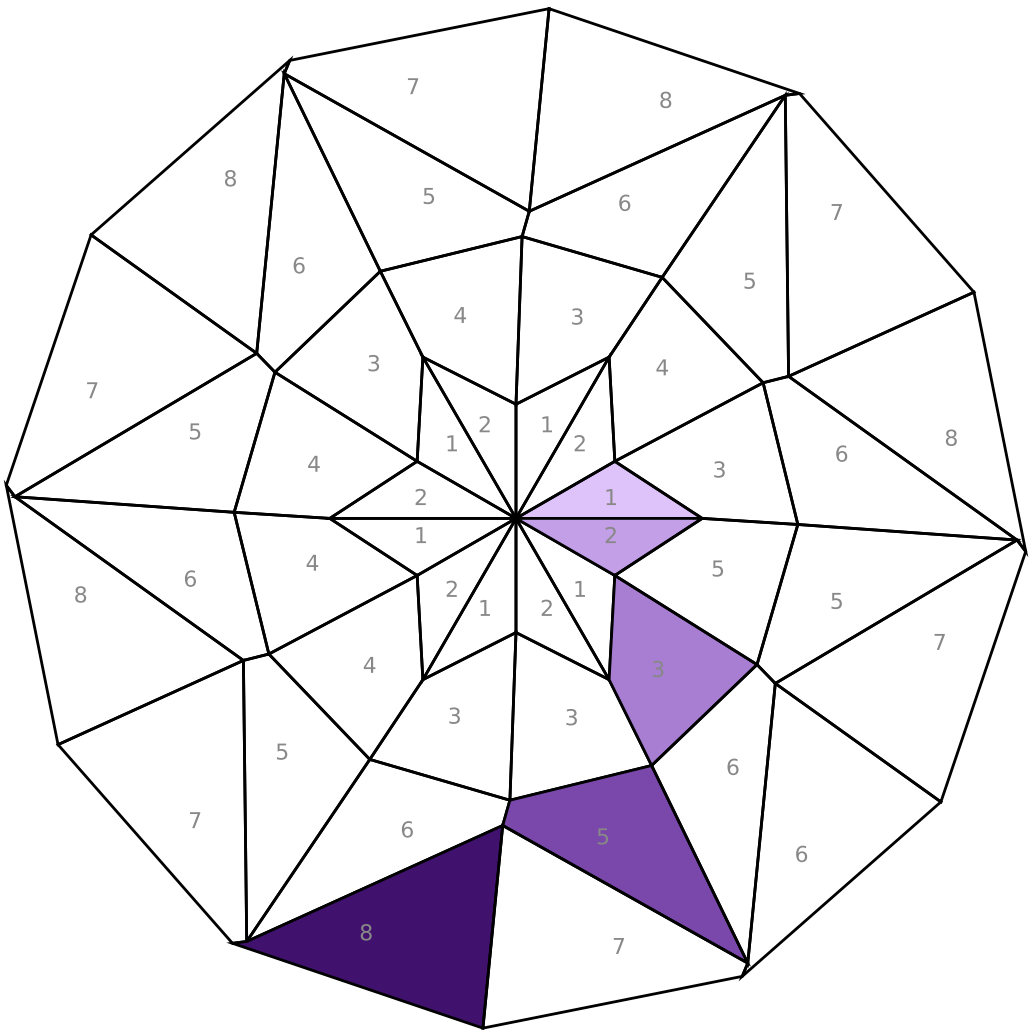
40. Rub el Hizb Star 5

1	2	3	4	5	6	7	8



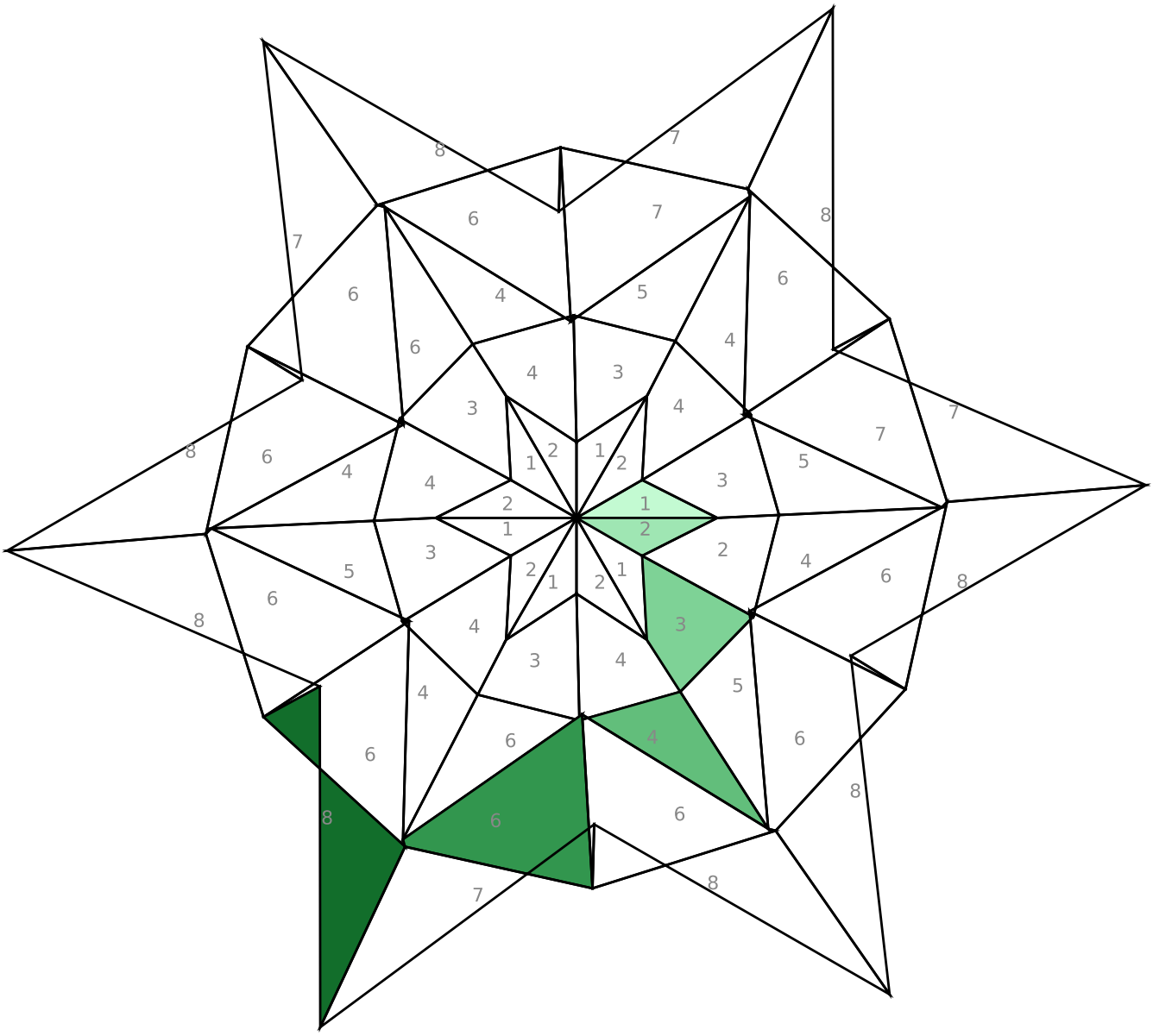
41. Khatam Weave 1

1	2	3	4	5	6	7	8



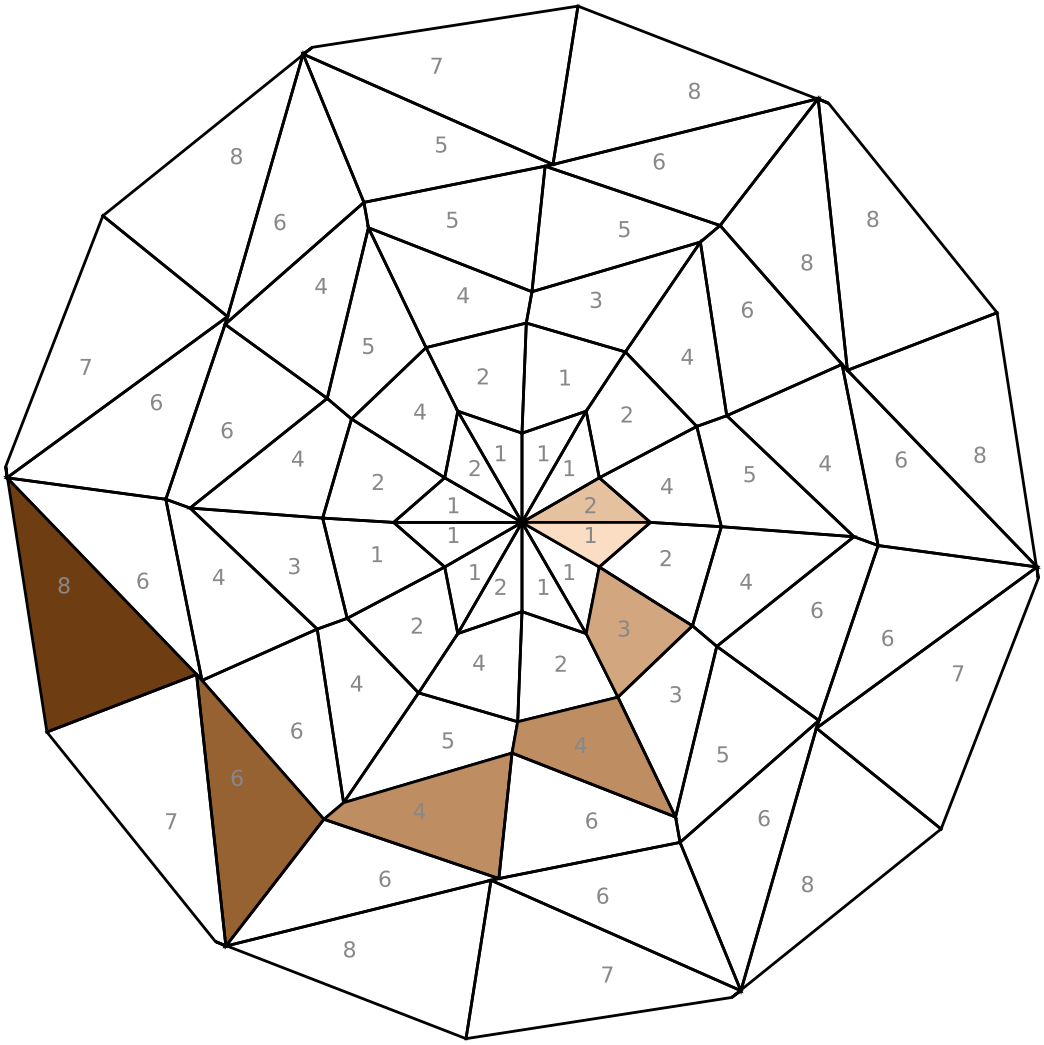
42. Khatam Weave 2

1	2	3	4	5	6	7	8



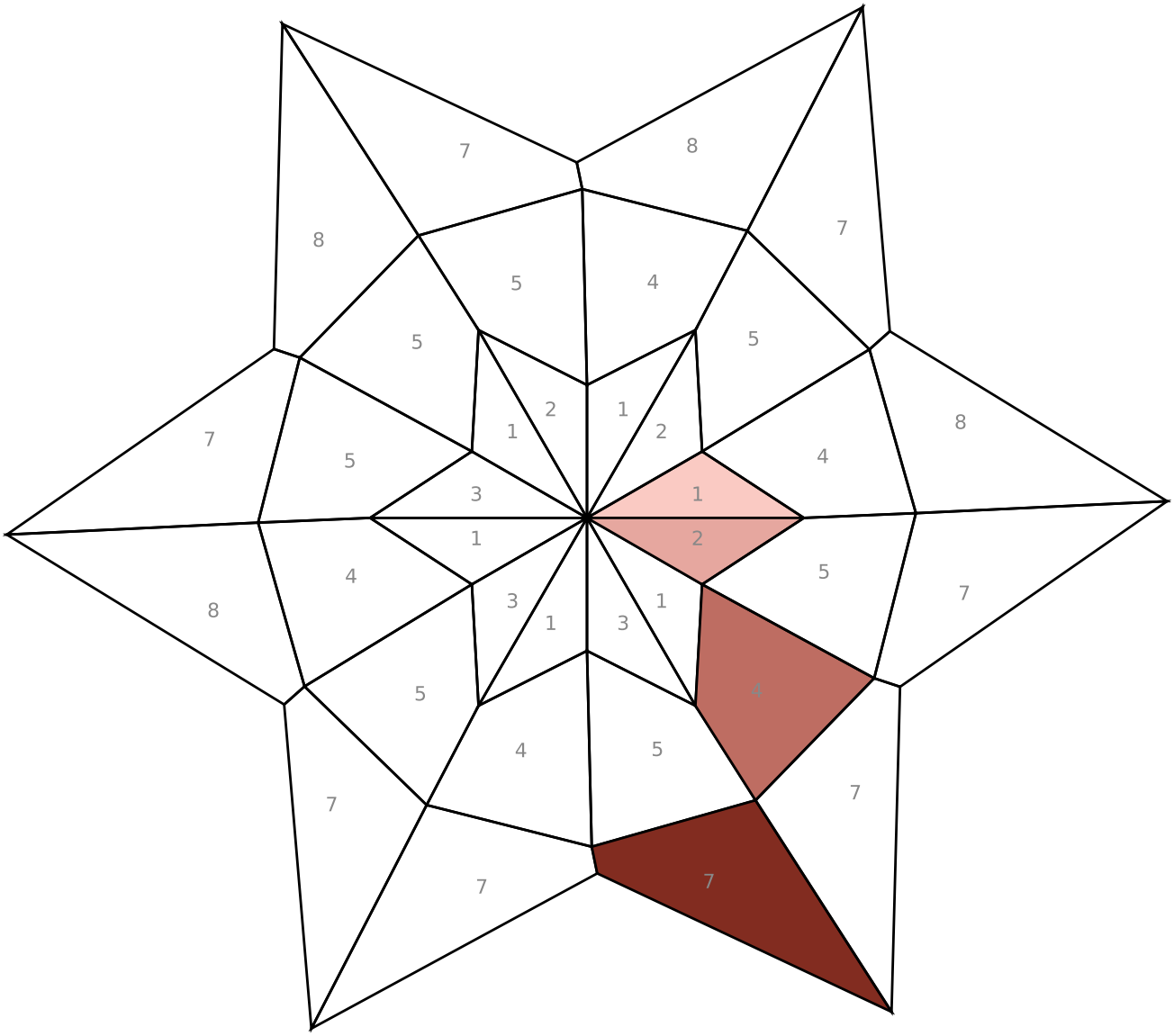
43. Khatam Weave 3

1	2	3	4	5	6	7	8



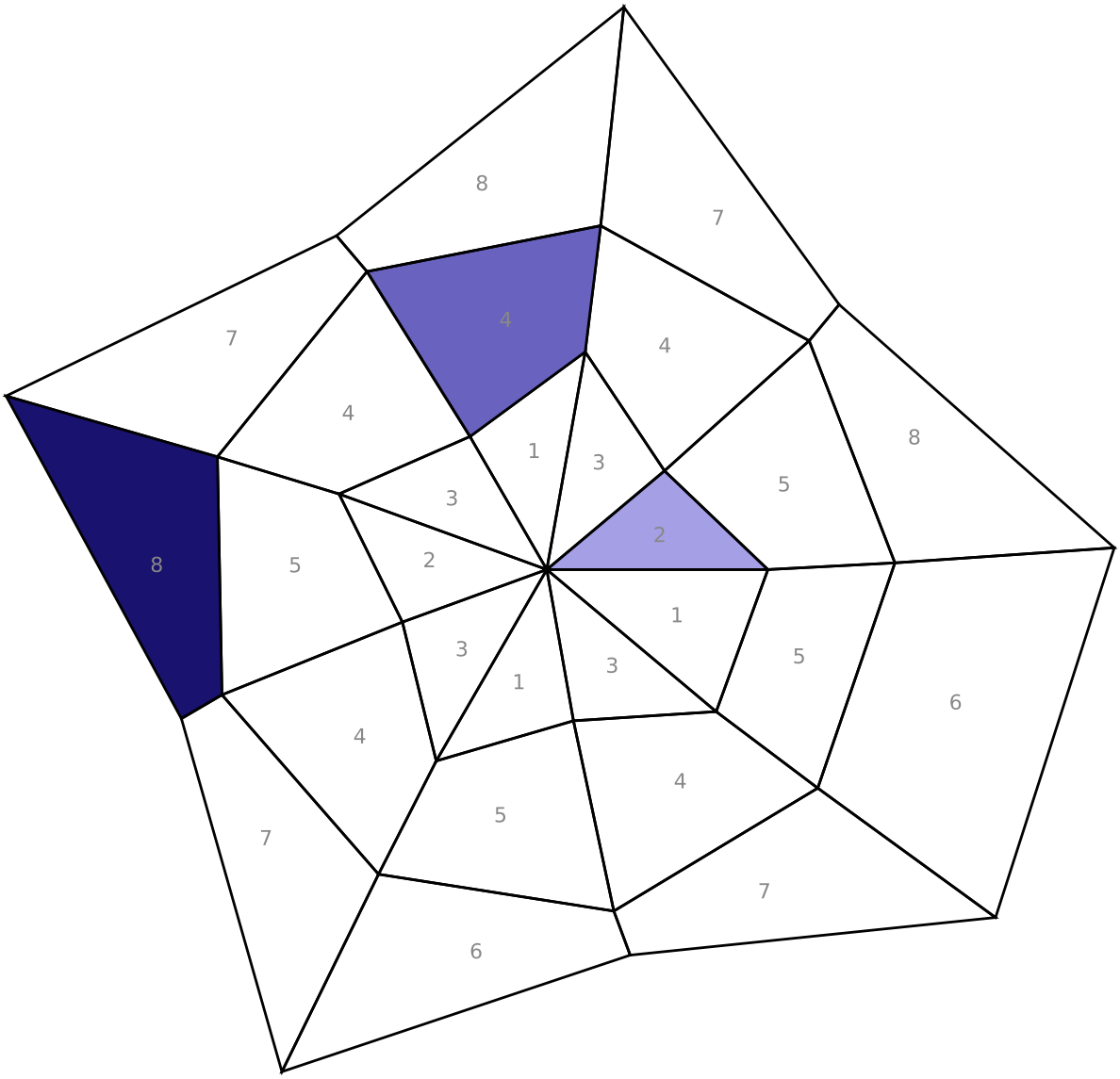
44. Khatam Weave 4

1	2	3	4	5	6	7	8



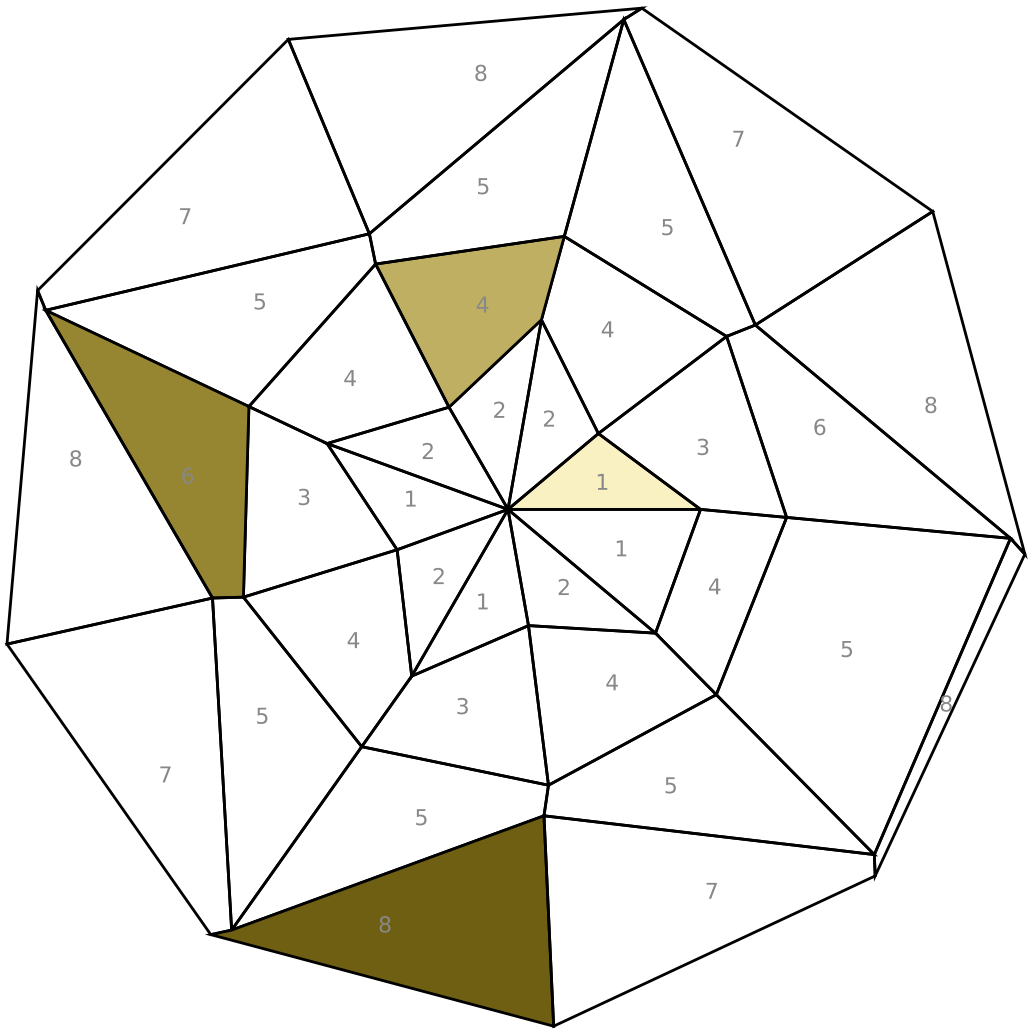
45. Khatam Weave 5

1	2	3	4	5	6	7	8



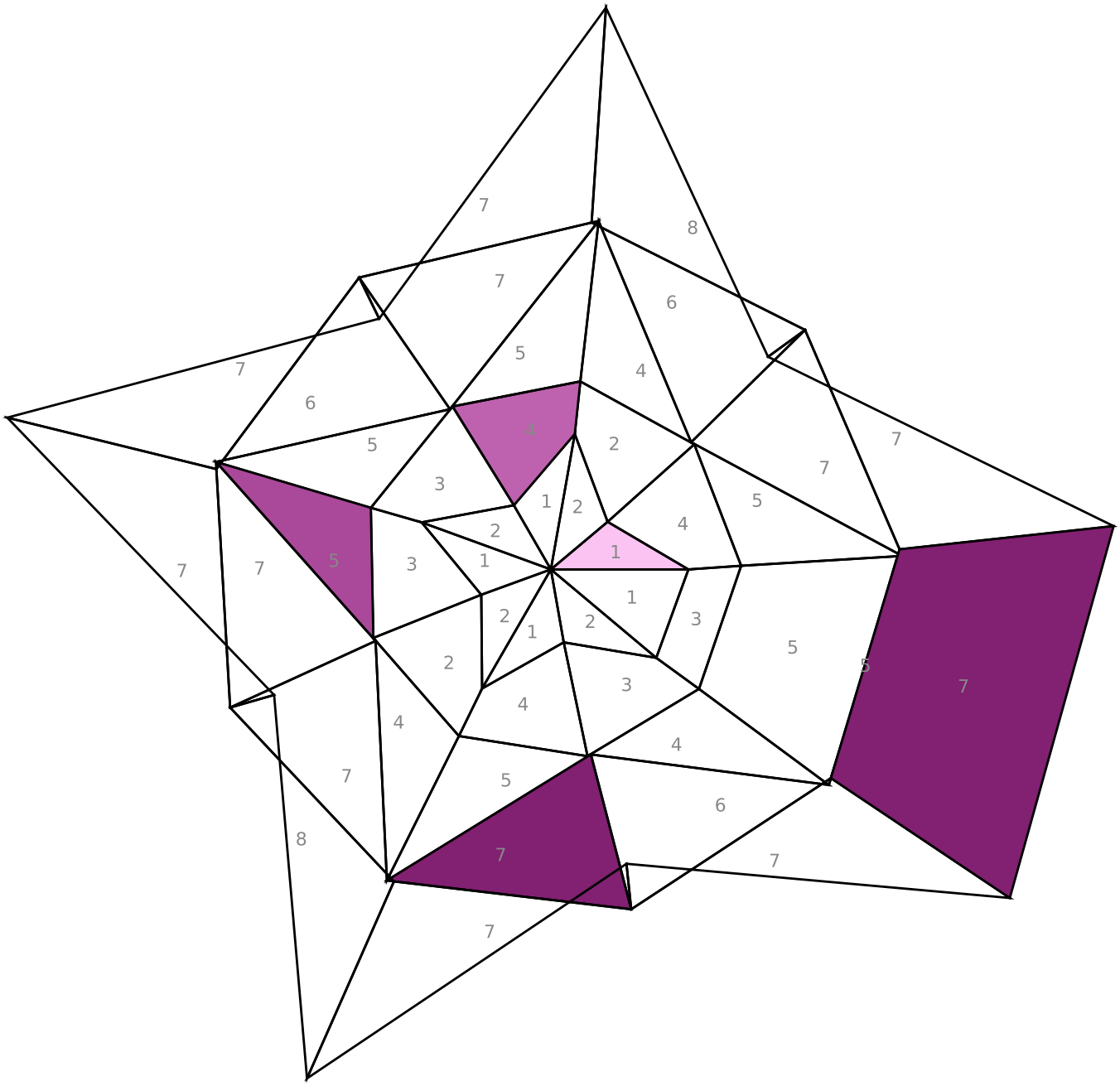
46. Isfahan Rosette 1

1	2	3	4	5	6	7	8



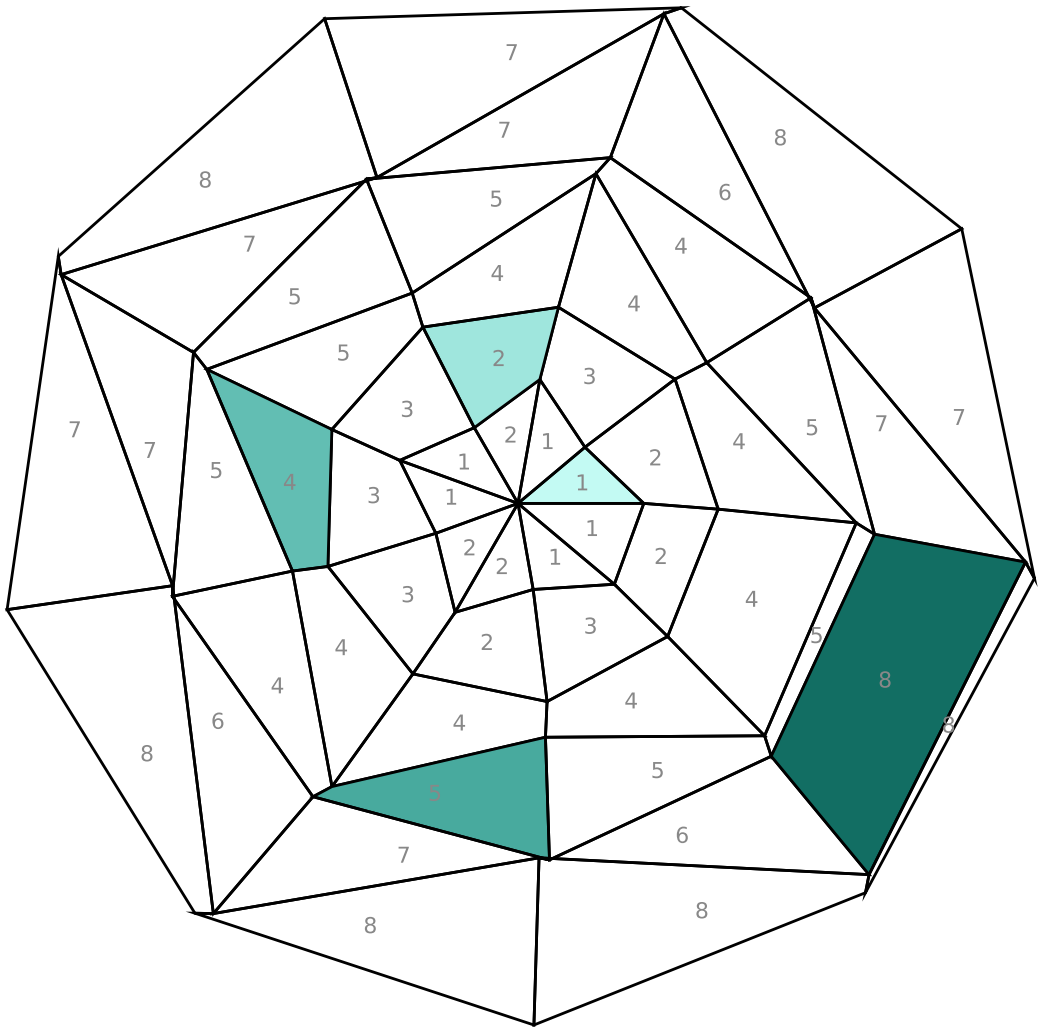
47. Isfahan Rosette 2

1	2	3	4	5	6	7	8



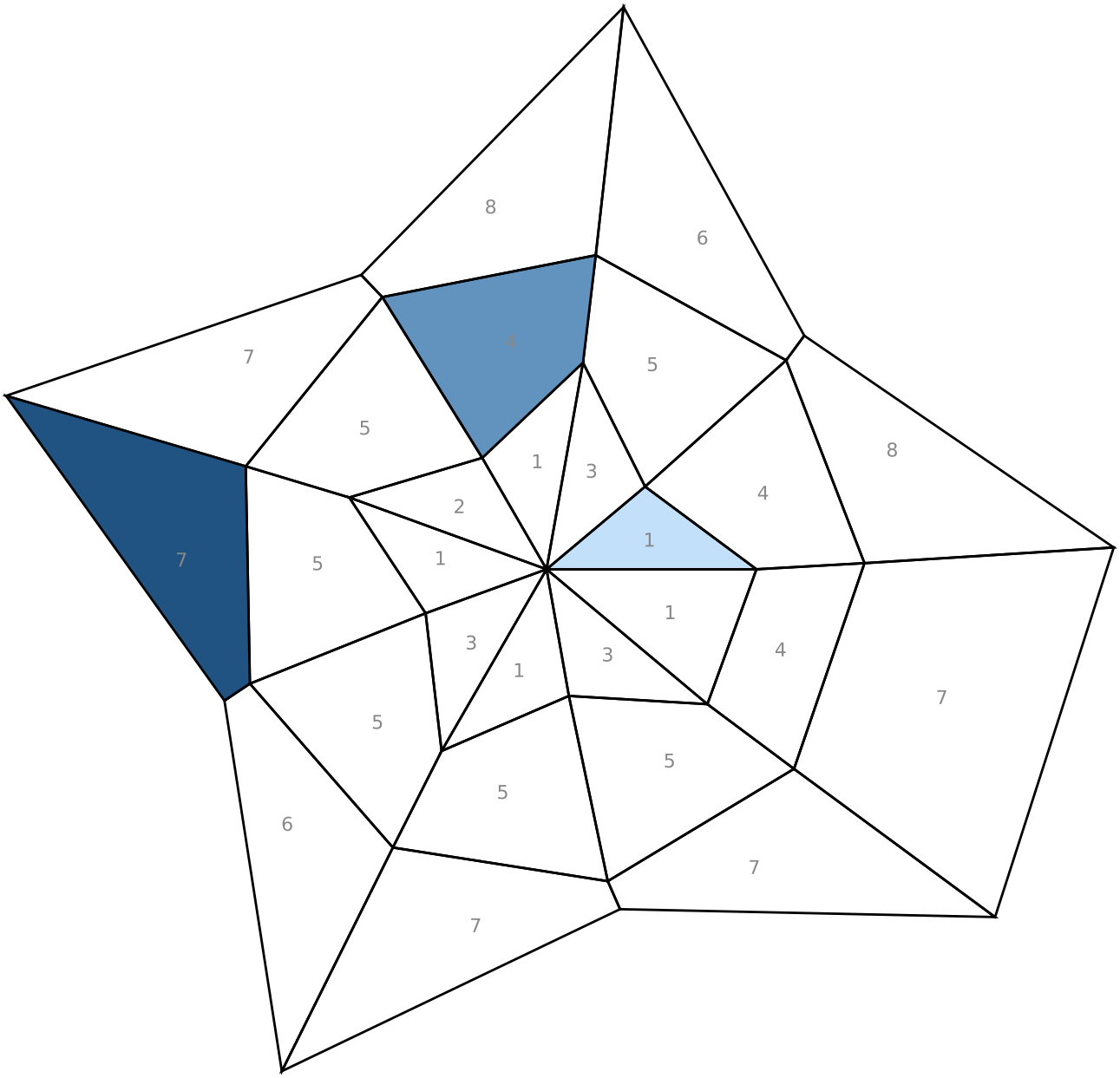
48. Isfahan Rosette 3

1	2	3	4	5	6	7	8



49. Isfahan Rosette 4

1	2	3	4	5	6	7	8



50. Isfahan Rosette 5

1	2	3	4	5	6	7	8